Chemotaxis models with signal-dependent sensitivity and a logistic-type source, III: Bifurcation and spike-like solutions

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Abstract

This is the third part of a series of papers on boundedness, global existence, and asymptotic behavior of positive classical solutions to the following parabolic–elliptic chemotaxis model,

$$\begin{cases} u_t = \Delta u - \chi_0 \nabla \cdot \left(\dfrac{u^m}{(1+v)^\beta} \nabla v \right) + au - bu^{1+\alpha}, & x \in \Omega, \\ 0 = \Delta v - \mu v + \nu u^\gamma, & x \in \Omega, \\ \dfrac{\partial u}{\partial n} = \dfrac{\partial v}{\partial n} = 0, & x \in \partial\Omega, \end{cases} \tag{CM}$$

where Ω is a bounded smooth domain in $\mathbb{R}^N$, and the parameters $\alpha, \gamma, m, \mu, \nu$ are positive, $\chi_0 \in \mathbb{R}$, and a, b, β are nonnegative. In the first part of this series [3], we identified several biologically relevant parameter regimes that guarantee the boundedness and global existence of positive classical solutions. In the second part [4], we investigated the persistence and stabilization of globally defined bounded positive solutions, with particular emphasis on quantifying the effects of the signal-dependence exponent β and the sign of the sensitivity coefficient χ_0 on the long-time dynamics. In this third part, treating χ_0 as the bifurcation parameter, we study bifurcation of positive constant equilibria of (CM) when the space dimension $N = 1$. We pay special attention to the effects of β and the domain size on the bifurcation.

By applying center manifold reduction, we derive a quantity β_{n_0} at a simple bifurcation mode n_0 of a positive constant solution (u^*, v^*), and show that super-critical (resp. sub-critical) pitchfork bifurcation occurs if $\beta_{n_0} > 0$ (resp. $\beta_{n_0} < 0$) when χ_0 passes through the bifurcation threshold χ^*. We find two interesting ways in which β enters: it shifts the bifurcation *threshold*, namely, larger signal-dependence exponent β needs proportionately larger χ_0 for the constant equilibrium to lose linear stability, and it also selects bifurcation *type*, namely, when β is small, the bifurcation may be super-critical, but when β is sufficiently large, the bifurcation is always sub-critical. Biologically, β thus acts twice in the same direction: it makes patterns harder to trigger, and it makes them more likely to appear abruptly rather than to grow continuously from the constant state. We also find some interesting effects of the domain size. For example, it is found that, in the non-minimal model, the bifurcation mode n_0 becomes larger and larger when the domain size increases and hence a large domain creates more complicated bifurcation patterns. In addition, under suitable conditions we prove unbounded continuation of the local bifurcation branches and prove that large-sensitivity stationary sequences satisfying a peak-gap hypothesis develop spike-like solutions. We give some numerical analysis of the continuation of local bifurcation and spike-like solutions. Our numerical time-evolution experiments suggest unbounded continuation of local bifurcation branches and increasingly sharp stationary profiles.

Keywords. Pitchfork bifurcation, center-manifold reduction, global bifurcation, signal-dependent sensitivity, fixed-mass chemotaxis, spike-like solution.

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1 Introduction

This paper is the third of a series of papers on the boundedness, global existence and asymptotic behavior of classical solutions to chemotaxis models of the following form on a bounded, connected, and smooth domain $\Omega \subset \mathbb{R}^N$:

$$\begin{cases} u_t = \Delta u - \nabla \cdot (u^m \chi(v) \nabla v) + au - bu^{1+\alpha}, & x \in \Omega, \\ 0 = \Delta v - \mu v + \nu u^\gamma, & x \in \Omega, \\ \dfrac{\partial u}{\partial n} = \dfrac{\partial v}{\partial n} = 0, & x \in \partial\Omega. \end{cases} \tag{1.1}$$

Here, the function $\chi(v)$, known in the literature as the *chemotactic sensitivity function*, is given by

$$\chi(v) = \frac{\chi_0}{(1+v)^\beta}. \tag{1.2}$$

In equations (1.1) and (1.2), m, α, μ, ν, and γ are positive constants, χ_0 is a real number, and a, b, and β are nonnegative constants.

System (1.1) describes the evolution of a biological species influenced by a chemical substance or signal produced by the species itself. In this context, $u(t,x)$ represents the population density of the biological species, and $v(t,x)$ represents the density of the chemical substance or signal. When $\beta = 0$, $\chi(v)$ simplifies to the constant chemotactic sensitivity function $\chi(v) \equiv \chi_0$. When $\beta > 0$, the magnitude of $\chi(v)$ decreases with the signal concentration. Such signal dependence can model partial loss of the signal through binding; see [32]. Related models with signal-dependent sensitivity or motility are studied in [1, 2, 28]. System (1.1) with $\beta > 0$ is known as a chemotaxis system with signal-dependent sensitivity. The coefficient $\chi_0 > 0$ indicates that the signal is a *chemoattractant* signal, while $\chi_0 < 0$ indicates that the signal is *chemorepellent*. The reaction term $au - bu^{1+\alpha}$ governs the local dynamics of the biological species and is known as a *logistic source*. When $a = b = 0$, (1.1) is a minimal chemotaxis system. In the literature, a chemotaxis system is also referred to as a *Keller–Segel system*, named after the pioneering works of E. F. Keller and L. A. Segel [13–15].

Central problems in studying (1.1) include determining whether solutions with given initial conditions exist globally and understanding the asymptotic behavior of globally defined bounded solutions of (1.1). For constant sensitivity, global existence, boundedness, and blow-up have been studied in parabolic–elliptic models; see, e.g., [7, 11, 23, 24, 27, 33]. Herrero, Medina, and Velázquez [9] considered a related parabolic–elliptic reduction. Herrero and Velázquez [10] studied finite-time singularity patterns in a related two-dimensional parabolic–elliptic reduction.

By contrast, fewer works have focused on the global existence and asymptotic behavior of positive solutions of (1.1). The objective of this series of papers is to investigate the global existence, boundedness, and asymptotic behavior of positive classical solutions of (1.1). In Part I [3], we studied the boundedness and global existence of positive classical solutions of (1.1) by examining the roles of negative chemotactic sensitivity χ_0, the nonlinear chemotactic mobility $\frac{u^m}{(1+v)^\beta}$, and the logistic-type source $u(a - bu^\alpha)$. In Part II [4], we studied orbitwise uniform persistence of globally defined, bounded, and positive solutions and stability of positive constant solutions under proper conditions on the parameters. Among other things, it is proved that the relevant positive constant solution loses stability when χ_0 passes through the corresponding critical sensitivity (see Proposition 1.1 for a little more detail).

In this paper, we explore the bifurcation of constant solutions of (1.1) when the domain is one-dimensional, that is, $\Omega = (0, L)$ for some $L > 0$, and $\chi(v)$ is given in (1.2). In this case, the system (1.1) can be written as

$$\begin{cases} u_t = u_{xx} - \chi_0\big(\dfrac{u^m}{(1+v)^\beta} v_x\big)_x + au - bu^{1+\alpha}, & x \in (0, L), \\ 0 = v_{xx} - \mu v + \nu u^\gamma, & x \in (0, L), \\ u_x(t,x) = v_x(t,x) = 0, & x = 0, L. \end{cases} \tag{1.3}$$

We also study spike-like stationary solutions of (1.3) when $\chi_0 \to \infty$.

We point out that there are several works on the bifurcation of positive constant solutions and the existence of spiky stationary solutions of (1.1) with various choices of $m, \alpha, \mu, \nu, \gamma$, and $\chi(v)$. For instance, for quadratic logistic growth, Wang, Yan, and Gai [29] studied a general one-dimensional stationary Keller–Segel system, deriving local bifurcation values and positive branches, global continuation properties, and possible boundary-spike limits along positive monotone solution sequences. Jin, Wang, and Zhang [12] treated stationary bifurcation on multidimensional bounded domains under simple-mode and nonresonance hypotheses, and established a local wave-mode se-

lection principle for stable branches. The closest predecessor for the present signal-dependent setting is the work of Kurt, Shen, and Xue [19], which established local and global bifurcation results for stationary solutions in a chemotaxis system with singular sensitivity and logistic source and analyzed spike formation along the bifurcating branches. The reader is referred to the works [5, 16–18, 21, 30, 31, 34], for existence and stability results on spiky solutions in related Keller–Segel models. Earlier studies in the Keller–Segel literature also treated nonconstant stationary states and pattern formation from complementary viewpoints; see, for example, Lin, Ni, and Takagi [22] for large-amplitude stationary solutions and Kuto, Osaki, Sakurai, and Tsujikawa [20] for spatial pattern formation in a chemotaxis-diffusion-growth model. Recently, on compact metric graphs, including intervals as a special case, Shemtaga, Shen, and Sukhtaiev [25] identified stability thresholds and bifurcation points for logistic Keller–Segel models.

We study bifurcation of positive constant solutions of (1.3) with $m, \alpha, \mu, \nu, \gamma > 0$, $\beta \geq 0$, and $a, b > 0$ or $a = b = 0$. Hence, our study includes the non-minimal model with logistic type source as well as the minimal model. We treat χ_0 as the bifurcation parameter, and study which type of bifurcation occurs when a positive constant solution of (1.3) loses its stability both theoretically and numerically. We pay special attention to the effects of β on the bifurcation of positive constant solutions of (1.3). Theoretically, we apply the center-manifold reduction presented by Henry [8, Chapter 6] to compute the local amplitude equation and determine the bifurcation type. Crandall–Rabinowitz theory [6] provides an alternative local framework, while the global extension of local bifurcation uses Theorem 4.4 of the quasilinear elliptic bifurcation framework developed by Shi and Wang [26]. Numerically, we discretize the system in space by a finite-difference method with Neumann boundary conditions, which produces discrete analogues of the instability thresholds that converge to their continuum counterparts as the mesh is refined. The time-evolution experiments illustrate the predicted local bifurcation directions as well as global extension of local bifurcation pattern; see Section 6.

In the following, we give a brief review on some relevant results obtained in Parts I and II of this series of papers, and then state the main results of the current paper.

1.1 A brief review of relevant results from Parts I and II

In this subsection, we present some relevant results obtained in Parts I and II of this series of papers.

First, let us define $C^+([0, L])$ by

$$C^+([0, L]) := \{u \in C([0, L]) \mid \inf_{x\in[0,L]} u(x) > 0\}.$$

Then, by standard arguments, one can show the local existence of classical solutions of (1.3) on a maximal interval $[0, T_{\max})$ with initial condition $u(0, x; u_0) = u_0(x) \in C^+([0, L])$. For the precise definition of classical solutions of (1.3) or (1.1), we refer the reader to [3, Definition 1.1] in Part I. The reader is also referred to Part I [3] for biologically relevant parameter regimes that guarantee the boundedness and global existence of positive classical solutions.

Next, note that when $a = b = 0$, for any positive constant $u^* > 0$, (u^*, v^*) is a positive constant solution of (1.3), where $v^* = \frac{\nu}{\mu}(u^*)^\gamma$. Hence, (1.3) with $a = b = 0$ has a one-parameter family of positive constant solutions, $\{(u^*, v^*)\}_{u^*>0}$. When $a, b > 0$, (1.3) has a unique positive constant solution (u^*, v^*) given by

$$(u^*, v^*) = \Big((\frac{a}{b})^{1/\alpha},\ \frac{\nu}{\mu}(\frac{a}{b})^{\gamma/\alpha} \Big). \tag{1.4}$$

Let (u^*, v^*) be a positive constant solution of (1.3). It is straightforward to verify that the linearization of (1.3) at (u^*, v^*) is given by

$$\begin{cases} u_t = u_{xx} - \chi_0 \dfrac{(u^*)^m}{(1+v^*)^\beta}\Big(\mu v - \nu\gamma (u^*)^{\gamma-1} u\Big) - a\alpha u, & x \in (0, L), \\ 0 = v_{xx} - \mu v + \nu\gamma (u^*)^{\gamma-1} u, & x \in (0, L), \\ u_x(t, x) = v_x(t, x) = 0, & x = 0, L. \end{cases} \tag{1.5}$$

Let $\{\lambda_n\}_{n=0}^{\infty}$ denote the eigenvalues of the negative Neumann Laplacian:

$$\begin{cases} -u_{xx} = \lambda u, & x \in (0, L), \\ \dfrac{\partial u}{\partial n} = 0, & x = 0, L \end{cases}$$

with $0 = \lambda_0 < \lambda_1 < \lambda_2 < \cdots$. Note that $\lambda_n = \frac{n^2\pi^2}{L^2}$. Let

$$\sigma_n = \sigma_n(\chi_0) := -\lambda_n + \chi_0 \nu\gamma \frac{(u^*)^{m+\gamma-1}}{(1+v^*)^\beta} \frac{\lambda_n}{\mu + \lambda_n} - a\alpha. \tag{1.6}$$

It can be verified directly that $\{\sigma_n\}_{n=0}^{\infty}$ are the eigenvalues of the following corresponding eigenvalue problem of (1.5),

$$\begin{cases} u_{xx} - \chi_0 \dfrac{(u^*)^m}{(1+v^*)^\beta}(\mu v - \nu\gamma (u^*)^{\gamma-1} u) - a\alpha u = \sigma u, & x \in (0, L), \\ 0 = v_{xx} - \mu v + \nu\gamma (u^*)^{\gamma-1} u, & x \in (0, L), \\ u_x(t, x) = v_x(t, x) = 0, & x = 0, L \end{cases} \tag{1.7}$$

and $\phi_n = \cos(\frac{n\pi x}{L})$ is an eigenfunction corresponding to the eigenvalue σ_n. Note that, when $a = b = 0$, $\sigma_0 = 0$, and when $a, b > 0$, $\sigma_0 < 0$.

For a given positive constant equilibrium (u^*, v^*) of (1.3), let

$$\chi^*(u^*) = \chi^*_{a,b,\beta}(u^*) := \inf_{n \geq 1} \Big[\frac{(1+v^*)^\beta}{\nu\gamma\,(u^*)^{m+\gamma-1}} \times \frac{(\lambda_n + a\alpha)(\mu + \lambda_n)}{\lambda_n}\Big], \tag{1.8}$$

where $v^* = \frac{\nu}{\mu}(u^*)^\gamma$. When $a, b > 0$, $u^* = (a/b)^{1/\alpha}$; when $a = b = 0$, $u^* > 0$ is free. Throughout the paper, $\chi^*(u^*)$ denotes $\chi^*_{a,b,\beta}(u^*)$ in the non-minimal case and $\chi^*_{0,0,\beta}(u^*)$ in the minimal case.

The following results, expressed in the present notation, were proved in [4].

Proposition 1.1 (Linear stability/instability)**.** *Let (u^*, v^*) be a positive constant solution of (1.3). If $\chi_0 < \chi^*(u^*)$, then (u^*, v^*) is linearly exponentially stable in the case $a, b > 0$, and is linearly exponentially stable with respect to perturbations with spatial average u^* in the case $a = b = 0$. If $\chi_0 > \chi^*(u^*)$, then (u^*, v^*) is linearly unstable.*

Proposition 1.1 identifies $\chi^*(u^*)$ as the boundary between the two strict linear regimes: the constant equilibrium is linearly stable below this threshold and linearly unstable above it, while the equality case is neutral at the linear level. It is therefore important to determine which nonconstant steady states bifurcate from (u^*, v^*) when χ_0 passes through $\chi^*(u^*)$ and how the parameters in (1.3) affect the bifurcation. In this paper, we study the bifurcation of constant solutions of (1.3) and provide both theoretical analysis and numerical analysis.

1.2 Statements of the main results and remarks

Let (u^*, v^*) be a positive constant solution of (1.3). By Proposition 1.1, it is expected that there are nonconstant steady states of (1.3) bifurcating from (u^*, v^*) when χ_0 passes through $\chi^*(u^*)$. To be more precise, let

$$\chi_n(u^*) = \frac{(1+v^*)^\beta}{\nu\gamma(u^*)^{m+\gamma-1}} \frac{(\lambda_n + a\alpha)(\mu+\lambda_n)}{\lambda_n}, \quad n \geq 1. \tag{1.9}$$

By (1.8) and (1.9), there is some $n_0 > 0$ independent of β and (u^*, v^*) such that

$$\chi^*(u^*) = \chi_{n_0}(u^*) \quad \forall\, \beta \geq 0. \tag{1.10}$$

Note that, if $\chi_0 - \chi^*(u^*) < 0$, then $\sigma_n(\chi_0) < 0$ for all $n \geq 1$, and if $0 < \chi_0 - \chi^*(u^*) \ll 1$, then $\sigma_{n_0}(\chi_0) > 0$ and $\sigma_n(\chi_0) < 0$ for $n \geq 1$ satisfying that $\chi_n(u^*) \neq \chi^*(u^*)$, where $\sigma_n(\cdot)$ is as in (1.6). Therefore, we call n_0 a *bifurcation mode* if (1.10) holds.

Let

$$L_0 := 0 \quad \text{and} \quad L_k = L_k(a) := \pi\sqrt{k(k+1)}/(a\alpha\mu)^{1/4} \text{ for } k \geq 1. \tag{1.11}$$

When $a = 0$, we agree on

$$L_1 = L_1(a) := \infty.$$

If $a > 0$, then $n_0 = k$ is the unique bifurcation mode when $L \in (L_{k-1}(a), L_k(a))$, and $n_0 = k, k+1$ are two bifurcation modes when $L = L_k(a)$ $(k \geq 1)$. If $a = 0$, then $n_0 = 1$ is the unique bifurcation mode for any $L \in (0, L_1(0)) = (0, \infty)$. Therefore, we say (L_{k-1}, L_k) is an *admissible interval for the domain size* L and we say L *is admissible* if $L \in (L_{k-1}, L_k)$ for some $k \geq 1$. Note that, in the minimal model, we have only one admissible interval $(0, L_1(0)) = (0, \infty)$ and any $L > 0$ is admissible.

Observe that if $n_0 \geq 1$ is a unique bifurcation mode, then

$$\sigma_{n_0}(\chi^*(u^*)) = 0 \quad \text{and} \quad \sigma_n(\chi^*(u^*)) < 0 \quad \forall\, \beta \geq 0,\ n \geq 1, n \neq n_0. \tag{1.12}$$

In addition, $\sigma_0 = -a\alpha < 0$ in the non-minimal model; in the minimal model, the neutral constant mode is excluded by the fixed-mass constraint. Hence, by Proposition 1.1, when χ_0 passes through $\chi^*(u^*)$ for any $\beta \geq 0$, the constant solution (u^*, v^*) of (1.3) loses its linear stability along the eigen-direction ϕ_{n_0}, and it is expected that there are non-constant solutions bifurcating from (u^*, v^*) along this eigen-direction.

Our first main theorem is on the local bifurcation of a positive constant solution (u^*, v^*) of (1.3), which will be proved in Section 2.

Theorem 1.1 (Local pitchfork bifurcation)**.** *Let (u^*, v^*) be a positive constant solution of (1.3). Assume that $n_0 \geq 1$ is a unique bifurcation mode. Let*

$$\alpha_{n_0} = \alpha_{n_0}(u^*, m, \beta, \gamma, \mu, \nu) := \partial_{\chi_0}\sigma_{n_0}(\chi^*(u^*)) = \nu\gamma \frac{(u^*)^{m+\gamma-1}}{(1+v^*)^\beta} \frac{\lambda_{n_0}}{\mu+\lambda_{n_0}} \,(>0). \tag{1.13}$$

There is a number β_{n_0} depending on u^* and all the parameters in the model (see (2.45)) such that the following hold.

If $\beta_{n_0} \neq 0$, pitchfork bifurcation occurs near (u^*, v^*) in (1.3) when χ_0 passes through χ^*. More precisely, there are two continuous curves of stationary solutions $\Gamma^\pm_{n_0}(\chi_0)$,

$$\Gamma^\pm_{n_0} : I_\varepsilon \longrightarrow C^2(\overline{\Omega}) \times C^2(\overline{\Omega}),$$

bifurcating from (u^*, v^*), *where* $I_\varepsilon := [\chi^*(u^*), \chi^*(u^*) + \varepsilon)$ *if* $\beta_{n_0} > 0$ *and* $I_\varepsilon := (\chi^*(u^*) - \varepsilon, \chi^*(u^*)]$ *if* $\beta_{n_0} < 0$ *for some* $\varepsilon > 0$, $\Gamma^{\pm}_{n_0}(\chi^*) = (u^*, v^*)$, $\Gamma^{+}_{n_0}(\chi_0) \neq \Gamma^{-}_{n_0}(\chi_0)$ *for* $\chi_0 \neq \chi^*$, *and the solutions on* $\Gamma^{\pm}_{n_0}$ *are nonconstant for* $\chi_0 \neq \chi^*(u^*)$.

Moreover, for $(u^{\pm}, v^{\pm}) = \Gamma^{\pm}_{n_0}(\chi_0)$ $(\chi_0 \in I_\varepsilon)$, *there hold*

$$u^{\pm} = u^* \pm \sqrt{\frac{\alpha_{n_0}}{\beta_{n_0}}(\chi_0 - \chi^*)}\phi_{n_0} + \mathcal{O}(|\chi_0 - \chi^*|),$$

and

$$v^{\pm} = v^* \pm \frac{\nu\gamma(u^*)^{\gamma-1}}{\lambda_{n_0} + \mu}\sqrt{\frac{\alpha_{n_0}}{\beta_{n_0}}(\chi_0 - \chi^*)}\phi_{n_0} + \mathcal{O}(|\chi_0 - \chi^*|).$$

If $\beta_{n_0} > 0$, *the bifurcation is super-critical, and for each fixed* $\chi_0 > \chi^*(u^*)$ *sufficiently close to* $\chi^*(u^*)$, $(u^{\pm}, v^{\pm}) \in \Gamma^{\pm}_{n_0}(\chi_0) \setminus \{(u^*, v^*)\}$ *are linearly exponentially stable in the case* $a, b > 0$, *and they are linearly exponentially stable with respect to perturbations with spatial average* u^* *in the case* $a = b = 0$.

If $\beta_{n_0} < 0$, *the bifurcation is sub-critical, and for each fixed* $\chi_0 < \chi^*(u^*)$ *sufficiently close to* $\chi^*(u^*)$, $(u^{\pm}, v^{\pm}) \in \Gamma^{\pm}_{n_0}(\chi_0) \setminus \{(u^*, v^*)\}$ *are linearly unstable.*

Before we state our next result, let us make a remark on Theorem 1.1.

Remark 1.1. Let (u^*, v^*) be a positive constant solution of (1.3) and $n_0 \geq 1$ be a unique bifurcation mode.

(1) β_{n_0} stated in Theorem 1.1 is derived explicitly via center manifold theory. Roughly speaking, let

$$u(t, x) = \tilde{u}(t, x) + u^*, \quad v(t, x) = \tilde{v}(t, x) + v^*,$$

and

$$\tilde{u}(t, x) = \sum_{n=0}^{\infty} u_n(t) \cos\left(\tfrac{n\pi x}{L}\right) \quad \text{and} \quad \tilde{v}(t, x) = \sum_{n=0}^{\infty} v_n(t) \cos\left(\tfrac{n\pi x}{L}\right).$$

Then by center manifold theory, the study of bifurcation solutions of (1.3) near (u^*, v^*) when χ_0 passes through χ^* is reduced to the study of bifurcation solutions of the following scalar ODE near 0 when $\tilde{\chi} = \chi_0 - \chi^*(u^*)$ passes through 0,

$$\frac{du_{n_0}}{dt} = \alpha_{n_0}\,\tilde{\chi}\,u_{n_0} - \beta_{n_0}\,u_{n_0}^3 + \mathcal{O}\big(|u_{n_0}|^2|\tilde{\chi}|^2 + |u_{n_0}|^3|\tilde{\chi}| + |u_{n_0}|^4\big)$$

(see (2.43)), where β_{n_0} is explicitly given in (2.45). It is then seen that the bifurcation is pitchfork as long as $\beta_{n_0} \neq 0$, and the sign of β_{n_0} determines the bifurcation direction or bifurcation type, i.e., the bifurcation is super-critical or sub-critical. Although using center-manifold reduction to derive this coefficient is standard, it is nontrivial and worth deriving the explicit formula for the coefficient β_{n_0}. Such a formula allows us to determine the effects of the parameters on the bifurcation (see Theorem 1.2 about the effects of β and L), which is important biologically.

(2) It is seen from Theorem 1.1 that there are non-constant positive stationary solutions (u, v) of (1.3) near (u^*, v^*) (with the average of u being u^* in the minimal model) for χ_0 near $\chi^*(u^*)$. By Proposition 1.1 and stable manifold theory (see [8, Theorem 5.1.1]), there are no non-constant positive stationary solutions (u, v) of (1.3) near (u^*, v^*) (with the average of u being u^* in the minimal model) for χ_0 near 0. When $\chi_0 = 0$, there are no any non-constant positive stationary solution (u, v) of (1.3) (with the average of u being u^* in the minimal model). This follows from the maximum principle when $a, b > 0$, and from $u_{xx} = 0$ with Neumann boundary conditions when $a = b = 0$.

Our next main theorem concerns the effects of β and L on the bifurcation of a positive constant solution (u^*, v^*) of (1.3), which will be proved in Section 3.

Theorem 1.2 (Effects of β and L on the bifurcation)**.** *Let (u^*, v^*) be a positive constant solution of (1.3). Assume that $n_0 \geq 1$ is a unique bifurcation mode. Let β_{n_0} be as in Theorem 1.1. Then the following hold.*

(1) Fix all the parameters except β. Then

$$\beta_{n_0} = \beta_{n_0}(\beta) = A\beta^2 + B\beta + C \tag{1.14}$$

for some constants A, B, C independent of β with $A < 0$ (see (3.4)). Hence, one of the following occurs when β changes:

(i) There is $\beta_+ > 0$ such that $\beta_{n_0}(\beta) > 0$ for $0 \leq \beta < \beta_+$, and $\beta_{n_0}(\beta) < 0$ for $\beta > \beta_+$.

(ii) There are $0 \leq \beta_- < \beta_+$ such that $\beta_{n_0}(\beta) < 0$ for $\beta \in [0, \beta_-)$, $\beta_{n_0}(\beta) > 0$ for $\beta \in (\beta_-, \beta_+)$, and $\beta_{n_0}(\beta) < 0$ for $\beta > \beta_+$.

(iii) There is $\beta_0 \geq 0$ such that $\beta_{n_0}(\beta_0) = 0$ and $\beta_{n_0}(\beta) < 0$ for all $\beta \geq 0$ with $\beta \neq \beta_0$.

(iv) $\beta_{n_0}(\beta) < 0$ for all $\beta \geq 0$.

(2) Fix all parameters except L. Consider $L \in (L_{k-1}(a), L_k(a))$ for some $k \geq 1$. Then $n_0 = k$ and

$$\beta_{n_0} = \frac{P(\lambda_{n_0})}{Q(\lambda_{n_0})}, \quad \lambda_{n_0} = \frac{\pi^2 n_0^2}{L^2}. \tag{1.15}$$

Here, in the non-minimal model,

$$\lambda_{n_0} \in \Big(\frac{k}{k+1}\sqrt{a\alpha\mu}, \frac{k}{k-1}\sqrt{a\alpha\mu}\Big) \quad \text{if } k \geq 2, \qquad \lambda_{n_0} \in \Big(\frac{1}{2}\sqrt{a\alpha\mu}, \infty\Big) \quad \text{if } k = 1,$$

P is a polynomial of degree at most five, and

$$Q(\lambda) = 12K^2(\mu+\lambda)^2(4\lambda^2 - a\alpha\mu) > 0 \quad \text{for } \lambda > \frac{\sqrt{a\alpha\mu}}{2} \quad \text{with} \quad K = \tfrac{\nu\gamma(u^*)^{m+\gamma-1}}{(1+v^*)^\beta}.$$

In the minimal model, $\lambda_{n_0} \in (0, \infty)$, P is a polynomial of degree at most three, and

$$Q(\lambda) = 24(u^*)^2(\mu+\lambda)^2 > 0 \qquad \text{for } \lambda > 0.$$

Moreover, there are constants c and β_∞, independent of L, such that

$$\beta_{n_0} = c\frac{\pi^2}{L^2} + \mathcal{O}(1) \quad \text{as } L \to 0+, \quad \text{and} \quad \beta_{n_0} \longrightarrow \beta_\infty \quad \text{as } L \to \infty \text{ admissible lengths.}$$

Hence, the following hold.

(i) Suppose $P \not\equiv 0$. When L changes in the admissible interval $(L_{k-1}(a), L_k(a))$, in the non-minimal model, the bifurcation type changes at most five times, and in the minimal model, it changes at most three times.

(ii) If $\beta_\infty > 0$ (respectively, $\beta_\infty < 0$), the bifurcation is super-critical (respectively, sub-critical) for every sufficiently large admissible length.

(iii) If $c > 0$ (respectively, $c < 0$), the bifurcation is super-critical (respectively, sub-critical) for all sufficiently small $L > 0$.

Before stating our next result, we make the following remark on the effects of β and L on the bifurcation of constant solutions.

Remark 1.2. Let (u^*, v^*) be a positive constant solution of (1.3).

(1) Assume that $n_0 \geq 1$ is a unique bifurcation mode. With all parameters except β fixed, $\beta_{n_0}(\beta)$ has at most two nonnegative zeros. At these values, higher-order terms are needed for bifurcation analysis; we do not analyze these cases. Put $\chi^*(u^*, \beta) = \chi^*(u^*)$. Theorems 1.1 and 1.2(1) reveal the following two effects of β:

- Threshold: $\chi^*(u^*, \beta)$ grows proportionally to $(1 + v^*)^\beta$ by (1.8), so large signal-dependence exponent β requires a larger χ_0 for non-constant stationary solutions to bifurcate from (u^*, v^*).

- Type: $\beta_{n_0}(\beta) = A\beta^2 + B\beta + C$ determines whether the bifurcation is super-critical (positive) or sub-critical (negative). Since $A < 0$, sufficiently large β gives sub-critical bifurcation. These local nonconstant branches lie below the bifurcation threshold and are unstable near the constant state.

(2) Fix all parameters except for L. In the non-minimal model, when $L = L_k$ $(k \geq 1)$, there are two bifurcation modes $n_0 = k$ and $k + 1$. In this case, the positive constant solution (u^*, v^*) loses its stability when χ_0 passes through χ^* along two eigen-directions ϕ_k and ϕ_{k+1}. We do not investigate bifurcation solutions in this case in this paper. With the other parameters fixed except L, Theorem 1.2(2) reveals the following effects of L:

- In the non-minimal model, when L increases along admissible lengths, the bifurcation threshold χ^* stays bounded, but bifurcation mode n_0 becomes larger and larger. Hence, a larger domain creates more complicated patterns.

- The sign of the coefficient c in Theorem 1.2(2) determines the bifurcation type for $0 < L \ll 1$. In the non-minimal model, by some direct calculation, we have

$$c = \frac{\mathsf{P} - 6\beta\gamma D v^*/(1 + v^*)}{24(u^*)^2} \qquad \text{and} \qquad \mathsf{P} = h(6\alpha - 4\gamma - m + 13), \tag{1.16}$$

where $h = \gamma + m - 1$ and $D = \alpha + 2 - \gamma$. Suppose $D > 0$ and $\mathsf{P} \geq 0$. Let $\beta_{\rm sd} = \frac{\mathsf{P}(1+v^*)}{6\gamma D v^*}$. Then

$$c = \frac{\gamma D v^*}{4(u^*)^2(1 + v^*)}(\beta_{\rm sd} - \beta). \tag{1.17}$$

Thus, as $L \to 0+$, $\beta_{n_0} \to +\infty$ for $0 \leq \beta < \beta_{\rm sd}$, so the small-domain bifurcation is super-critical, and $\beta_{n_0} \to -\infty$ for $\beta > \beta_{\rm sd}$, so the small-domain bifurcation is sub-critical. If instead $D > 0$ and $\mathsf{P} < 0$, then $c < 0$ for every $\beta \geq 0$, so the small-domain bifurcation is sub-critical.

- The sign of β_∞ determines the bifurcation type for large admissible L. For the non-minimal model, as $L \to \infty$, we have

$$\beta_\infty = A_\infty\beta^2 + B_\infty\beta + C_\infty, \qquad A_\infty < 0,$$

where the coefficients are independent of β. Indeed, this follows by evaluating the quadratic formula for $\beta_{n_0}(\beta)$ at $\lambda = \sqrt{a\alpha\mu}$. It follows that for relatively large β, the bifurcation type for large admissible L is always sub-critical. But for relatively small β, the bifurcation type for large admissible L may be super-critical if $C_\infty > 0$.

We remark that the local bifurcation branches $\Gamma^{\pm}_{n_0}$ established in Theorem 1.1 can be extended for χ_0 not near χ^* by applying global bifurcation theory in the literature, for example, [26, Theorem 4.4]. To be more precise, let

$$X \coloneqq \begin{cases} \left\{ u \in H^2((0,L)) \,\middle|\, u_x(0) = u_x(L) = 0 \right\}, & \text{if} \quad a, b > 0 \\ \left\{ u \in H^2((0,L)) \,\middle|\, u_x(0) = u_x(L) = 0, \ \int_0^L u(x)dx = 0 \right\}, & \text{if} \quad a = b = 0 \end{cases} \tag{1.18}$$

with norm $\|u\| = \|u\|_{H^2((0,L))}$. For a given positive constant solution (u^*, v^*) of (1.3), let

$$\mathcal{A} = \mathcal{A}(u^*) := \{ w + u^* \,|\, w \in X, \ w(x) + u^* > 0 \quad \text{for} \quad x \in [0, L] \}, \tag{1.19}$$

n_0 be a bifurcation mode, and

$$W \coloneqq \left\{ w \in X : \int_0^L w(x)\phi_{n_0}(x)\, dx = 0 \right\}.$$

Since $\phi_{n_0} \in X$ and $\int_0^L \phi_{n_0}^2\, dx = L/2$, this is a closed complement of $\operatorname{span}\{\phi_{n_0}\}$ in X.

Let $\Gamma^{\pm}_{n_0}(\chi_0) = (u^{\pm}(\chi_0), v^{\pm}(\chi_0))$, and define the punctured local half-curves

$$\mathcal{G}^{\pm}_{n_0} \coloneqq \left\{ (\chi_0, u^{\pm}(\chi_0)) : \chi_0 \in I_\varepsilon \setminus \{\chi^*\} \right\}.$$

Here I_ε and $\Gamma^{\pm}_{n_0}$ are as in Theorem 1.1, with $\varepsilon > 0$ chosen sufficiently small. Equip $(0,\infty) \times \mathcal{A}$ with the topology induced by $\mathbb{R} \times H^2(0, L)$, and let

$$\mathcal{S} = \{ (\chi_0, u) \in (0,\infty) \times \mathcal{A} : u \neq u^*, \ (u, v) \text{ solves the stationary system for some } v \}.$$

Let $\mathcal{C}$ be the connected component of $\overline{\mathcal{S}}$ containing (χ^*, u^*), where the closure is taken in $\mathbb{R} \times \mathcal{A}$. By Remark 1.1(2),

$$\overline{\mathcal{S}} \subset (0,\infty) \times \mathcal{A}. \tag{1.20}$$

Then we have the following theorem, which will be proved in Section 4.

Theorem 1.3 (Extension of the local bifurcation branches)**.** *Let (u^*, v^*) be a positive constant solution of (1.3). Assume that n_0 is a unique bifurcation mode, and that the coefficient $\beta_{n_0} \neq 0$.*

Then the following hold.

(1) Let $\mathcal{C}^{\pm}_{n_0}$ be the connected component of $\mathcal{C} \setminus \mathcal{G}^{\mp}_{n_0}$ containing $\mathcal{G}^{\pm}_{n_0}$. Each of these sets contains (χ^*, u^*) and is relatively closed in $(0,\infty) \times \mathcal{A}$. Moreover, each of the sets $\mathcal{C}^{+}_{n_0}$ and $\mathcal{C}^{-}_{n_0}$ satisfies one of the following alternatives:

(i) $\mathcal{C}^{\pm}_{n_0}$ is not compact in $(0,\infty) \times \mathcal{A}$;

(ii) $\mathcal{C}^{\pm}_{n_0}$ contains another point $(\widehat{\chi}_0, u^*)$ with $\widehat{\chi}_0 \neq \chi^*(u^*)$;

(iii) $\mathcal{C}^{\pm}_{n_0}$ contains a point $(\widehat{\chi}_0, u^* + w)$ with $w \in W \setminus \{0\}$.

(2) Assume $a, b > 0$, $0 < \gamma \leq 1 + \alpha$, and $1 \leq m \leq \alpha + 2$. If $\mathcal{C}^{+}_{n_0}$ (respectively, $\mathcal{C}^{-}_{n_0}$) is not compact, then

$$\sup\left\{ \chi_0 : (\chi_0, u) \in \mathcal{C}^{+}_{n_0} \right\} = +\infty \quad \text{(respectively,} \quad \sup\left\{ \chi_0 : (\chi_0, u) \in \mathcal{C}^{-}_{n_0} \right\} = +\infty). \tag{1.21}$$

(3) *Assume* $a = b = 0$, $1 \le m \le 2$, *and* $0 < \gamma \le 1$. *Then neither* $\mathcal{C}_1^+$ *nor* $\mathcal{C}_1^-$ *satisfies alternative (ii) or (iii) in (1), and hence both sets are not compact and* (1.21) *holds. Moreover, every nonconstant solution on* $\mathcal{C}_1^+$ *(respectively,* $\mathcal{C}_1^-$*) is strictly decreasing (respectively, strictly increasing) on* $[0, L]$. *For any sequence* $(\chi_j, u_j) \in \mathcal{C}_1^+$ *(respectively,* $\mathcal{C}_1^-$*) with* $\chi_j \to \infty$,

$$\int_0^L f(x)u_j(x)\,dx \longrightarrow Lu^* f(0) \qquad (\textit{respectively, } Lu^* f(L))$$

for every $f \in C([0, L])$. *In particular,* $\max_{x\in[0,L]} u_j(x) \to \infty$.

We remark that Theorem 1.3(3) indicates the monotone steady states in $\mathcal{C}_1^+$ (resp. $\mathcal{C}_1^-$) form a spike as $\chi_0 \to \infty$. In general, we have the following theorem concerning positive nonconstant stationary solutions of (1.3) for large χ_0, which is proved in Section 5.

Theorem 1.4 (Spike-like stationary solutions)**.** *Let* (u^*, v^*) *be a positive constant solution of* (1.3) *and* (u_n, v_n) *be positive nonconstant classical stationary solutions of* (1.3) *with* $\chi_0 = \chi_n \to \infty$ *and* $\int_0^L u_n\,dx = Lu^*$ *for every* n *in the case that* $a = b = 0$. *If*

$$\liminf_{n\to\infty} \max_{[0,L]} u_n > u^*,$$

then there is a subsequence $\{u_{n_k}\}$ *and a point* $x^* \in [0, L]$ *such that, for every* $\delta > 0$,

$$\max_{[x^*-\delta,x^*+\delta]\cap[0,L]} |(u_{n_k})_x| \longrightarrow \infty \quad \text{as} \quad n_k \to \infty. \tag{1.22}$$

Moreover, in the non-minimal model, if $0 < \gamma < 1 + \alpha$, *then*

$$u_{n_k}^\gamma \rightharpoonup c \ \ in \ \ L^q, \quad v_{n_k} \to \frac{\nu}{\mu}c \ \ in \ \ C^1, \quad v_{n_k} \rightharpoonup \frac{\nu}{\mu}c \ \ in \ \ W^{2,q} \tag{1.23}$$

for $q = \frac{1+\alpha}{\gamma}$ *and some constant* $0 \le c \le (u^*)^\gamma$.

We make the following remark on Theorem 1.4 before stating our last result.

Remark 1.3. (1) Theorem 1.4 applies to a branch from Theorem 1.3(1) if it extends to arbitrarily large χ_0 and $\liminf_{\chi_0\to\infty} \max_{[0,L]} u(\cdot;\chi_0) > u^*$. In the non-minimal model, whether there are positive non-constant stationary solutions satisfying the conditions in Theorem 1.4 remains open. Our numerical experiments presented in Section 6 provide numerical evidence consistent with such behavior.

(2) For the non-minimal model, the spikes in Theorem 1.4 cannot carry a Dirac mass for any $\alpha, \gamma > 0$. Indeed, the stationary logistic identity $a\int_0^L u_n\,dx = b\int_0^L u_n^{1+\alpha}\,dx$ gives $\int_0^L u_n^{1+\alpha}\,dx < L\,(u^*)^{1+\alpha}$ by Lemma 4.1(1). Hölder's inequality therefore yields

$$\int_{(x^*-\delta,x^*+\delta)\cap(0,L)} u_n\,dx \le (2\delta)^{\frac{\alpha}{1+\alpha}} \|u_n\|_{L^{1+\alpha}} < u^* L^{\frac{1}{1+\alpha}} (2\delta)^{\frac{\alpha}{1+\alpha}}.$$

Thus, the mass in shrinking neighborhoods tends to zero uniformly in n, even though the gradients diverge.

(3) For nonconstant solutions $(\chi_n, u_n) \in \mathcal{C}_1^\pm$ under the assumptions of Theorem 1.3(3), $\chi_n \to \infty$ implies $\max_{[0,L]} u_n \longrightarrow \infty$. Thus, the liminf condition in Theorem 1.4 is automatic for these sequences.

The theoretical results stated in the above provide several insights into the bifurcation of positive constant solutions of (1.3), including the finding of the number β_{n_0} whose sign determines the local bifurcation type, the effects of β and L on the bifurcation, the extension of local bifurcation branches, and existence of spike-like solutions. Theorem 1.3(3) proves endpoint Dirac-mass concentration for the specified minimal-model first-mode branches. The existence and limiting behavior of spike-like solutions outside these hypotheses remain questions for further study. In this paper, we investigate large-sensitivity profiles numerically.

To carry out numerical analysis of the bifurcation and large-sensitivity profiles, we discretize the space variable in (1.3) by a finite-difference method on N uniform subintervals, with $N+1$ grid points, to obtain a system of $N+1$ coupled ordinary differential equations (see (6.1) for the non-minimal model and (6.2) for the minimal model). Let (u^*, v^*) be a positive constant equilibrium of (1.3). For each $N \geq 1$, set $h = L/N$. Note that the eigenvalues of the discrete Laplacian $-\Delta_h$ are given by

$$\lambda_{0,N}^{\mathrm{disc}} = 0, \qquad \lambda_{n,N}^{\mathrm{disc}} = \frac{4}{h^2}\sin^2\left(\frac{n\pi}{2N}\right), \qquad 1 \leq n \leq N. \tag{1.24}$$

It is not difficult to see that the eigenvalues of the linearization of the semidiscrete system (6.1) at (u^*, v^*) are

$$\sigma_{n,N}^{\mathrm{disc}}(\chi_0) := -\lambda_{n,N}^{\mathrm{disc}} - a\alpha + \frac{\chi_0 \nu\gamma (u^*)^{m+\gamma-1}}{(1+v^*)^{\beta}} \frac{\lambda_{n,N}^{\mathrm{disc}}}{\mu + \lambda_{n,N}^{\mathrm{disc}}}, \qquad 0 \leq n \leq N. \tag{1.25}$$

In particular, $\sigma_{0,N}^{\mathrm{disc}} = -a\alpha$. For each $1 \leq n \leq N$, the equation $\sigma_{n,N}^{\mathrm{disc}}(\chi_0) = 0$ determines the instability threshold for that mode. The discrete instability threshold is therefore

$$\chi_N^{*,\mathrm{disc}}(u^*, \beta) := \min_{1\leq n\leq N} \frac{(1+v^*)^{\beta}}{\nu\gamma (u^*)^{m+\gamma-1}} \frac{\left(\lambda_{n,N}^{\mathrm{disc}} + a\alpha\right)\left(\mu + \lambda_{n,N}^{\mathrm{disc}}\right)}{\lambda_{n,N}^{\mathrm{disc}}}. \tag{1.26}$$

In the minimal case, this threshold is understood on the fixed-mass subspace, so that the neutral constant mode is excluded. The following proposition, proved in Subsection 6.1, establishes convergence to the continuous instability threshold.

Proposition 1.2 (Consistency of thresholds)**.** *Fix $m, \alpha, \mu, \nu, \gamma > 0$ and $\beta \geq 0$, and assume either $a, b > 0$ or $a = b = 0$. Then*

$$\lim_{N\to\infty} \chi_N^{*,\mathrm{disc}}(u^*, \beta) = \chi_{a,b,\beta}^*(u^*). \tag{1.27}$$

We carry out several numerical experiments using the above mentioned numerical scheme. Our experiments suggest that nonconstant profiles persist as χ_0 increases and that their maxima remain separated from the equilibrium value u^* in the tested regimes (see Figures 6.1–6.4), where $u^* = (a/b)^{1/\alpha}$ for the non-minimal model and is the prescribed average for the minimal model.

The rest of the paper is organized as follows. In Section 2, we prove Theorem 1.1. In Section 3, we prove Theorem 1.2. In Section 4, we prove Theorem 1.3. We prove Theorem 1.4 in Section 5. In Section 6, we establish the threshold consistency in Proposition 1.2 and present several numerical experiments.

2 Local bifurcation of constant solutions and proof of Theorem 1.1

In this section, we study bifurcation of positive constant solutions (u^*, v^*) of (1.3) with $a, b > 0$ or $a = b = 0$ and $\Omega = (0, L)$. We prove Theorem 1.1 in Subsections 2.1–2.2. We begin with a lemma collecting several technical trigonometric identities that will be used later in the proofs.

Lemma 2.1. *Let $(u_i)_{i\geq 0}$, $(v_j)_{j\geq 0}$, and $(w_\ell)_{\ell\geq 0}$ belong to $\ell^1(\mathbb{N}_0)$. We have*

$$\int_0^L \Big(\sum_{i=0}^{\infty} u_i \cos(\tfrac{i\pi x}{L})\Big) \sin(\tfrac{n\pi x}{L}) \sin(\tfrac{k\pi x}{L})\, dx = \frac{L}{4}\left(u_{|n-k|} - u_{n+k} + \delta_{nk} u_0\right), \qquad n,k \geq 1, \tag{2.1}$$

$$\begin{aligned}
&\int_0^L \Big(\sum_{i=0}^{\infty} u_i \cos(\tfrac{i\pi x}{L})\Big)\Big(\sum_{j=0}^{\infty} v_j \cos(\tfrac{j\pi x}{L})\Big) \sin(\tfrac{n\pi x}{L}) \sin(\tfrac{k\pi x}{L})\, dx \\
&\quad = \frac{L}{8}\Big(\sum_{i-j=n-k} u_i v_j + \sum_{i-j=-(n-k)} u_i v_j + \sum_{i+j=n-k} u_i v_j + \sum_{i+j=-(n-k)} u_i v_j \\
&\quad - \sum_{i-j=n+k} u_i v_j - \sum_{i-j=-(n+k)} u_i v_j - \sum_{i+j=n+k} u_i v_j \Big), \qquad n,k \geq 1,
\end{aligned} \tag{2.2}$$

$$\begin{aligned}
&\int_0^L \Big(\sum_{i=0}^{\infty} u_i \cos\left(\tfrac{i\pi x}{L}\right)\Big)\Big(\sum_{j=0}^{\infty} v_j \cos\left(\tfrac{j\pi x}{L}\right)\Big) \cos\left(\tfrac{k\pi x}{L}\right) dx \\
&\quad = \frac{L}{4}\Big(\sum_{i-j=k} u_i v_j \; + \sum_{i-j=-k} u_i v_j \; + \sum_{i+j=k} u_i v_j \Big) + \frac{L}{4}\mathbf{1}_{\{k=0\}} u_0 v_0, \qquad k \geq 0,
\end{aligned} \tag{2.3}$$

and

$$\begin{aligned}
&\int_0^L \Big(\sum_{i=0}^{\infty} u_i \cos\left(\tfrac{i\pi x}{L}\right)\Big)\Big(\sum_{j=0}^{\infty} v_j \cos\left(\tfrac{j\pi x}{L}\right)\Big)\Big(\sum_{\ell=0}^{\infty} w_\ell \cos\left(\tfrac{\ell\pi x}{L}\right)\Big) \cos\left(\tfrac{k\pi x}{L}\right) dx \qquad (2.4)\\
&=\frac{L}{8}\Big(\sum_{i+j+\ell=k} u_i v_j w_\ell + \sum_{i+j-\ell=k} u_i v_j w_\ell + \sum_{i+j-\ell=-k} u_i v_j w_\ell + \sum_{i-j+\ell=k} u_i v_j w_\ell \\
&\quad + \sum_{i-j+\ell=-k} u_i v_j w_\ell + \sum_{i-j-\ell=k} u_i v_j w_\ell + \sum_{i-j-\ell=-k} u_i v_j w_\ell \Big) + \frac{L}{8}\mathbf{1}_{\{k=0\}} u_0 v_0 w_0, \qquad k \geq 0. \qquad (2.5)
\end{aligned}$$

All indices in the sums range over $\mathbb{N}_0$, and a sum whose constraint has no solutions is understood to be zero.

Proof. The identities follow from the product-to-sum formulas and termwise integration. □

2.1 Local pitchfork bifurcation and proof of Theorem 1.1 for the non-minimal model

In this subsection, we prove Theorem 1.1 for the non-minimal model.

Proof of Theorem 1.1 for the non-minimal model. Our main idea is to reduce the bifurcation analysis of the PDE system (1.3) to a scalar ODE by applying center manifold theory.

For clarity, we divide the proof into five steps.

Step 1. In this step, we construct a center manifold of (1.3) at (u^*, v^*) for χ_0 near $\chi^*(u^*)$.

First, we make the following change of variables:

$$u(t,x) = \tilde{u}(t,x) + u^*, \quad v(t,x) = \tilde{v}(t,x) + v^*.$$

Then $(\tilde{u}, \tilde{v})$ satisfies

$$\tilde{u}_t = \tilde{u}_{xx} - \chi_0\Big[\Big(B_0 + B_u\tilde{u} + B_v\tilde{v} + B_{uu}\tilde{u}^2 + B_{uv}\tilde{u}\tilde{v} + B_{vv}\tilde{v}^2 + \mathcal{O}\big(|\tilde{u}|^3 + |\tilde{v}|^3\big) + \mathcal{O}\big(|\tilde{u}|^4 + |\tilde{v}|^4\big)\Big)\tilde{v}_x\Big]_x - b\Big(A'\tilde{u} + A''\tilde{u}^2 + A'''\tilde{u}^3 + \mathcal{O}(|\tilde{u}|^4)\Big), \tag{2.6}$$

and

$$0 = \tilde{v}_{xx} - \mu\tilde{v} + C'\tilde{u} + C''\tilde{u}^2 + C'''\tilde{u}^3 + \mathcal{O}(|\tilde{u}|^4), \tag{2.7}$$

where

$$B_0 \coloneqq \frac{(u^*)^m}{(1+v^*)^\beta}, \qquad B_u \coloneqq \frac{m(u^*)^{m-1}}{(1+v^*)^\beta}, \qquad B_v \coloneqq -\frac{\beta(u^*)^m}{(1+v^*)^{\beta+1}}, \tag{2.8}$$

$$B_{uu} \coloneqq \frac{m(m-1)(u^*)^{m-2}}{2(1+v^*)^\beta}, \qquad B_{uv} \coloneqq -\frac{m\beta(u^*)^{m-1}}{(1+v^*)^{\beta+1}}, \qquad B_{vv} \coloneqq \frac{\beta(\beta+1)(u^*)^m}{2(1+v^*)^{\beta+2}}, \tag{2.9}$$

$$A' \coloneqq \alpha(u^*)^\alpha, \qquad A'' \coloneqq \tfrac{1}{2}(1+\alpha)\alpha(u^*)^{\alpha-1}, \qquad A''' \coloneqq \tfrac{1}{6}(1+\alpha)\alpha(\alpha-1)(u^*)^{\alpha-2}, \tag{2.10}$$

and

$$C' \coloneqq \nu\gamma(u^*)^{\gamma-1}, \qquad C'' \coloneqq \frac{\nu\gamma(\gamma-1)(u^*)^{\gamma-2}}{2}, \qquad C''' \coloneqq \frac{\nu\gamma(\gamma-1)(\gamma-2)(u^*)^{\gamma-3}}{6}. \tag{2.11}$$

For undifferentiated Taylor terms, $\mathcal{O}(|\tilde{u}|^4)$ denotes a function-valued remainder $R_4(\tilde{u})(t,x)$ satisfying $|R_4(\tilde{u})(t,x)| \le C|\tilde{u}(t,x)|^4$ for all $x \in (0,L)$ whenever $\|\tilde{u}(t,\cdot)\|_{L^\infty(0,L)}$ is sufficiently small; the same convention applies to $\mathcal{O}(|\tilde{v}|^4)$, to mixed remainders such as $\mathcal{O}(|\tilde{u}|^4 + |\tilde{v}|^4)$, and $\mathcal{O}(|\tilde{u}|^3 + |\tilde{v}|^3)$.

Next, let

$$X = \Big\{w \in H^2(0,L) : w_x(0) = w_x(L) = 0\Big\},$$

and choose $\theta \in (3/4, 1)$. Let $A = I - \partial_{xx}$ on $L^2(0,L)$, with domain X, and define the fractional power space $X_\theta = D(A^\theta)$ with its graph norm. Then X_θ is embedded into $C^1([0,L])$ (see [8, Theorem 1.6.1]). Note that

$$\tilde{v} = \mathcal{K}(\tilde{u}),$$

where $\mathcal{K}$ is defined by

$$\Big(\mu - \partial_{xx}\Big)\mathcal{K}(w) = \nu\Big[\Big(u^* + w\Big)^\gamma - (u^*)^\gamma\Big], \qquad \partial_x\mathcal{K}(w) = 0 \quad \text{at } x = 0, L. \tag{2.12}$$

After substituting $\widetilde{v} = \mathcal{K}(\tilde{u})$ into (2.6), we get

$$\begin{aligned}\tilde{u}_t = \tilde{u}_{xx} - \chi_0\Big[&\Big(B_0 + B_u\tilde{u} + B_v\mathcal{K}(\tilde{u}) + B_{uu}\tilde{u}^2 + B_{uv}\tilde{u}\mathcal{K}(\tilde{u}) + B_{vv}\big(\mathcal{K}(\tilde{u})\big)^2 \\ &+\mathcal{O}\big(|\tilde{u}|^3 + |\mathcal{K}(\tilde{u})|^3\big) + \mathcal{O}\big(|\tilde{u}|^4 + |\mathcal{K}(\tilde{u})|^4\big)\Big)\Big(\mathcal{K}(\tilde{u})\Big)_x\Big]_x \\ &- b\Big(A'\tilde{u} + A''\tilde{u}^2 + A'''\tilde{u}^3 + \mathcal{O}(|\tilde{u}|^4)\Big).\end{aligned} \tag{2.13}$$

Now, we decompose X_θ as follows:

$$X_\theta = \tilde{X}_\theta \oplus \operatorname{span}\big(\phi_{n_0}\big),$$

that is, for any $u \in X_\theta$, there are unique $\tilde{u} \in \tilde{X}_\theta$ and $u_{n_0} \in \mathbb{R}$ such that

$$u = \tilde{u} + u_{n_0}\phi_{n_0},$$

where $\tilde{X}_\theta \subset X_\theta$, and for any $\tilde{u} \in \tilde{X}_\theta$, $\tilde{u}$ can be expanded as

$$\tilde{u} = \sum_{k \neq n_0} u_k \cos(\tfrac{k\pi x}{L}),\ \ u_k \in \mathbb{R}.$$

Let

$$\tilde{\chi} \coloneqq \chi_0 - \chi^*(u^*), \tag{2.14}$$

and consider (2.13) adjoined with the trivial equation

$$\frac{d\tilde{\chi}}{dt} = 0. \tag{2.15}$$

By applying center manifold theory (e.g. [8, Theorems 6.1.7 and 6.2.1]), there is a C^4 map $\tilde{h} : \mathrm{span}\{\phi_{n_0}\} \times \mathbb{R} \to \tilde{X}_\theta$ such that

$$\tilde{h}(0,0) = 0, \quad D\tilde{h}(0,0) = 0, \quad \tilde{h}(0,\tilde{\chi}) = 0 \quad \forall\, |\tilde{\chi}| \ll 1,$$

and

$$\mathcal{M} = \{\big(\tilde{h}(u_{n_0},\tilde{\chi}) + u_{n_0}\phi_{n_0}, \tilde{\chi}\big)\,|\,|u_{n_0}| \ll 1,\ |\tilde{\chi}| \ll 1\} \tag{2.16}$$

is locally invariant under (2.13)+(2.15), and local bifurcating solutions of (2.13)+(2.15) lie in this invariant manifold.

Step 2. For $(\tilde{u}_0, \tilde{\chi}) \in \mathcal{M}$, let $(\tilde{u}(t,x;\tilde{u}_0,\tilde{\chi}),\tilde{\chi})$ be the solution of (2.13)+(2.15) with $\tilde{u}(0,x;\tilde{u}_0,\tilde{\chi}) = \tilde{u}_0(x)$. We may put $\tilde{u}(t,x) = \tilde{u}(t,x;\tilde{u}_0,\tilde{\chi})$ if no confusion occurs. Let $\tilde{v}(t,\cdot) = \mathcal{K}(\tilde{u}(t,\cdot))$. Let

$$\tilde{u}(t,x) = \sum_{n=0}^{\infty} u_n(t)\cos\Big(\tfrac{n\pi x}{L}\Big) \quad \text{and} \quad \tilde{v}(t,x) = \sum_{n=0}^{\infty} v_n(t)\cos\Big(\tfrac{n\pi x}{L}\Big). \tag{2.17}$$

In this step, we derive the expressions for $v_k(t)$ in terms of u_n.

Note that

$$\mathcal{M} \subset X \times \mathbb{R}.$$

Hence, $\{u_n(t)\}, \{v_n(t)\}, \{\frac{n^2\pi^2}{L^2}u_n(t)\}, \{\frac{n^2\pi^2}{L^2}v_n(t)\} \in \ell^2$. Consequently, $\{u_n(t)\}, \{v_n(t)\} \in \ell^1$, and

$$\tilde{u}_x(t,x) = -\sum_{n=0}^{\infty} \tfrac{n\pi}{L}u_n(t)\sin(\tfrac{n\pi x}{L}) \qquad \text{and} \qquad \tilde{u}_{xx}(t,x) = -\sum_{n=0}^{\infty} \tfrac{n^2\pi^2}{L^2}u_n(t)\cos(\tfrac{n\pi x}{L}),$$

and similarly for $\tilde{v}_x$ and $\tilde{v}_{xx}$. From (2.7), we have

$$\begin{aligned} 0 = &-\sum_{n=0}^{\infty} \frac{n^2\pi^2}{L^2}v_n(t)\cos(\tfrac{n\pi x}{L}) - \mu\sum_{n=0}^{\infty} v_n(t)\cos(\tfrac{n\pi x}{L}) + C'\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L}) \\ &+ C''\Big(\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L})\Big)^2 + C'''\Big(\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L})\Big)^3 + \mathcal{O}\big(|\tilde{u}|^4\big). \end{aligned}$$

For $k=0$, integrating the above equation over $(0,L)$, we have

$$v_0 = \frac{C'}{\mu}u_0 + \frac{C''}{\mu}\Big(u_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}u_n^2\Big) + \frac{C'''}{\mu}\Big(u_0^3 + \frac{3}{2}u_0\sum_{n=1}^{\infty}u_n^2 + \frac{3}{4}\sum_{i,j\geq 1}u_iu_ju_{i+j}\Big) + \mathcal{O}_{v,0}\left(||u||_{\ell^1}^4\right), \quad (2.18)$$

Here, writing R_v for the Taylor remainder in (2.7), the scalar remainder is $(\mu L)^{-1}\int_0^L R_v\,dx$. For $k\geq 1$, we multiply by $\cos(k\pi x/L)$ and integrate over $(0,L)$. By Lemma 2.1,

$$\begin{aligned} v_k(t) = {} & C_k^{(1)}u_k(t) + C_k^{(2)}\Big(\sum_{i-j=k}u_iu_j + \sum_{i-j=-k}u_iu_j + \sum_{i+j=k}u_iu_j\Big) \\ & + C_k^{(3)}\Big(\sum_{i+j+\ell=k}u_iu_ju_\ell + \sum_{i+j-\ell=k}u_iu_ju_\ell + \sum_{i+j-\ell=-k}u_iu_ju_\ell + \sum_{i-j+\ell=k}u_iu_ju_\ell \\ & + \sum_{i-j+\ell=-k}u_iu_ju_\ell + \sum_{i-j-\ell=k}u_iu_ju_\ell + \sum_{i-j-\ell=-k}u_iu_ju_\ell\Big) + \mathcal{O}_{v,k}\Big(\|u\|_{\ell^1}^4\Big), \end{aligned} \quad (2.19)$$

where

$$C_k^{(1)} := \frac{C'}{\mu+\lambda_k}, \qquad C_k^{(2)} := \frac{C''}{2(\mu+\lambda_k)}, \qquad C_k^{(3)} := \frac{C'''}{4(\mu+\lambda_k)}, \quad k\geq 0. \quad (2.20)$$

For $k\geq 1$, the scalar remainder in (2.19) is $2\,[L(\mu+\lambda_k)]^{-1}\int_0^L R_v\cos(k\pi x/L)\,dx$. In the above, we write $u=(u_n)_{n\geq 0}$ and set $\|u\|_{\ell^1}=\sum_{n\geq 0}|u_n|$ and $\|u\|_{\ell^2}=\Big(\sum_{n\geq 0}|u_n|^2\Big)^{1/2}$. The notation $\mathcal{O}_{v,k}(\|u\|_{\ell^1}^j)$ and $\mathcal{O}_{u,k}(\|u\|_{\ell^1}^j)$ denotes scalar remainders in the respective mode equations, bounded by $C_k\|u\|_{\ell^1}^j$ for χ_0 in a fixed neighborhood of χ^*. Constants may depend on k; the signal remainder sequence is summable because of the factor $(\mu+\lambda_k)^{-1}$.

Step 3. In this step, we derive the system of first order ODEs satisfied by $\{u_n(t)\}_{n=0}^{\infty}$.

First, by (2.6), we have

$$\Big(\sum_{n=0}^{\infty}u_n(t)\cos(\tfrac{n\pi x}{L})\Big)_t = I_0+I_1+I_2+I_3+I_4+I_5+I_6+R_u, \quad (2.21)$$

Here $R_u = R_u^{(0)} + \partial_x R_u^{(1)}$, where $|R_u^{(0)}|\leq C|\tilde{u}|^4$ and $|R_u^{(1)}|\leq C\left(|\tilde{u}|+|\tilde{v}|\right)^3|\tilde{v}_x|$, with $R_u^{(1)}=0$ at the endpoints. The retained terms are

$$I_0 := -\sum_{n=0}^{\infty}\lambda_n\,u_n(t)\cos(\tfrac{n\pi x}{L}) - a\alpha\sum_{n=0}^{\infty}u_n(t)\cos(\tfrac{n\pi x}{L}),$$

$$I_1 := \chi_0\Big[\Big\{B_0 + B_u\sum_{n=0}^{\infty}u_n(t)\cos\left(\tfrac{n\pi x}{L}\right) + B_v\sum_{n=0}^{\infty}v_n(t)\cos\left(\tfrac{n\pi x}{L}\right)\Big\}\times\sum_{n=0}^{\infty}\tfrac{n\pi}{L}v_n(t)\sin\left(\tfrac{n\pi x}{L}\right)\Big]_x,$$

$$I_2 := \chi_0\Big[B_{uu}\Big(\sum_{n=0}^{\infty}u_n(t)\cos\left(\tfrac{n\pi x}{L}\right)\Big)^2\times\sum_{n=0}^{\infty}\tfrac{n\pi}{L}v_n(t)\sin\left(\tfrac{n\pi x}{L}\right)\Big]_x,$$

$$I_3 := \chi_0\Big[B_{uv}\Big(\sum_{n=0}^{\infty}u_n(t)\cos(\tfrac{n\pi x}{L})\Big)\Big(\sum_{n=0}^{\infty}v_n(t)\cos(\tfrac{n\pi x}{L})\Big)\times\sum_{n=0}^{\infty}\tfrac{n\pi}{L}v_n(t)\sin(\tfrac{n\pi x}{L})\Big]_x,$$

$$I_4 \coloneqq \chi_0\Big[B_{vv}\big(\sum_{n=0}^{\infty} v_n(t)\cos(\tfrac{n\pi x}{L})\big)^2 \times \sum_{n=0}^{\infty} \tfrac{n\pi}{L} v_n(t)\sin(\tfrac{n\pi x}{L})\Big]_x,$$

$$I_5 \coloneqq -bA''\Big(\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L})\Big)^2, \quad I_6 \coloneqq -bA'''\Big(\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L})\Big)^3.$$

Next, we project (2.21) onto the kth mode to get the ODE satisfied by $u_k(t)$. The regularity supplied by the parabolic equation gives for $k = 0, 1, 2, \cdots$,

$$\int_0^L \Big(\sum_{n=0}^{\infty} u_n(t)\cos(\tfrac{n\pi x}{L})\Big)_t \cos\big(\tfrac{k\pi x}{L}\big)dx = \int_0^L u_k'(t)\cos^2\big(\tfrac{k\pi x}{L}\big)dx = \begin{cases} Lu_k'(t) & \text{if} \quad k = 0, \\ \frac{L}{2}u_k'(t) & \text{if} \quad k \geq 1. \end{cases}$$

Hence,

$$u_k'(t)\int_0^L \cos^2\big(\tfrac{k\pi x}{L}\big)dx = \sum_{\ell=0}^{6} \mathcal{F}_k^{(\ell)} + \mathcal{O}_{u,k}(\|u\|_{\ell^1}^4) \quad \text{for} \quad k = 0, 1, 2, \cdots, \tag{2.22}$$

where

$$\mathcal{F}_k^{(\ell)} = \mathcal{F}_k^{(\ell)}(u, v) \coloneqq \int_0^L I_\ell(t, x)\cos\big(\tfrac{k\pi x}{L}\big)dx, \quad \ell = 0, 1, \ldots, 6, \ k = 0, 1, 2, \ldots. \tag{2.23}$$

The scalar remainder in (2.22) is $\int_0^L R_u \cos(k\pi x/L)\,dx$. Integrating its flux part by parts and using $\|\tilde{v}\|_{C^1} \leq C\|\tilde{u}\|_\infty \leq C\|u\|_{\ell^1}$ gives the stated fourth-order bound for each fixed k; division by L for $k = 0$ and by $L/2$ for $k \geq 1$ is absorbed in the mode-ODE remainders below.

For $k = 0$, Neumann boundary conditions imply $\int_0^L I_\ell(t, x)\,dx = 0$, $\ell = 1, 2, 3, 4$, and using the cosine orthogonality and Lemma 2.1 on the remaining integrals we get

$$\begin{aligned}\sum_{\ell=0}^{6} \mathcal{F}_0^{(\ell)}(u, v) =& -aL\alpha u_0(t) - bLA''\Big(u_0(t)^2 + \frac{1}{2}\sum_{n=1}^{\infty} u_n(t)^2\Big) \\ & - bLA'''\Big(\frac{1}{4}u_0(t)^3 + \frac{3}{4}\sum_{i,j\geq 0} u_i(t)u_j(t)u_{i+j}(t)\Big).\end{aligned}$$

For $k \geq 1$, we have

$$\mathcal{F}_k^{(0)} = \frac{L}{2}\Big(-\lambda_k + a - b(1+\alpha)(u^*)^\alpha\Big)u_k(t) = -\frac{L}{2}(\lambda_k + a\alpha)u_k(t),$$

$$\begin{aligned}\mathcal{F}_k^{(1)} =& \frac{\chi_0\pi^2}{4L}k^2 v_k\left(2B_0 + B_u u_0 + B_v v_0\right) \\ & + \frac{\chi_0\pi^2}{4L}k\sum_{n=1}^{\infty} n v_n\left[B_u\left(u_{|n-k|} - u_{n+k}\right) + B_v\left(v_{|n-k|} - v_{n+k}\right)\right].\end{aligned}$$

$$\begin{aligned}\mathcal{F}_k^{(2)} = \chi_0\,\frac{\pi^2}{8L}\,k\sum_{n=1}^{\infty} n\,v_n(t)\,B_{uu}\Big(&\sum_{i-j=n-k} u_i u_j + \sum_{i-j=-(n-k)} u_i u_j \\ &+ \sum_{i+j=n-k} u_i u_j + \sum_{i+j=-(n-k)} u_i u_j - \sum_{i-j=n+k} u_i u_j - \sum_{i-j=-(n+k)} u_i u_j - \sum_{i+j=n+k} u_i u_j\Big),\end{aligned}$$

$$\mathcal{F}_k^{(3)} = \chi_0 \frac{\pi^2}{8L} k \sum_{n=1}^{\infty} n\, v_n(t)\, B_{uv}\Big(\sum_{i-j=n-k} u_i v_j + \sum_{i-j=-(n-k)} u_i v_j + \sum_{i+j=n-k} u_i v_j + \sum_{i+j=-(n-k)} u_i v_j - \sum_{i-j=n+k} u_i v_j - \sum_{i-j=-(n+k)} u_i v_j - \sum_{i+j=n+k} u_i v_j \Big),$$

$$\mathcal{F}_k^{(4)} = \chi_0 \frac{\pi^2}{8L} k \sum_{n=1}^{\infty} n\, v_n(t)\, B_{vv}\Big(\sum_{i-j=n-k} v_i v_j + \sum_{i-j=-(n-k)} v_i v_j + \sum_{i+j=n-k} v_i v_j + \sum_{i+j=-(n-k)} v_i v_j - \sum_{i-j=n+k} v_i v_j - \sum_{i-j=-(n+k)} v_i v_j - \sum_{i+j=n+k} v_i v_j \Big),$$

$$\mathcal{F}_k^{(5)} = -bA'' \frac{L}{4}\Big(\sum_{i-j=k} u_i u_j + \sum_{i-j=-k} u_i u_j + \sum_{i+j=k} u_i u_j \Big),$$

and

$$\mathcal{F}_k^{(6)} = -bA''' \frac{L}{8}\Big(\sum_{i+j+\ell=k} u_i u_j u_\ell + \sum_{i+j-\ell=k} u_i u_j u_\ell + \sum_{i+j-\ell=-k} u_i u_j u_\ell + \sum_{i-j+\ell=k} u_i u_j u_\ell + \sum_{i-j+\ell=-k} u_i u_j u_\ell + \sum_{i-j-\ell=k} u_i u_j u_\ell + \sum_{i-j-\ell=-k} u_i u_j u_\ell \Big).$$

By (2.22) and (2.23), we then have

$$\frac{du_0}{dt} = -a\alpha\, u_0 - bA''\Big(u_0^2 + \frac{1}{2}\sum_{i=1}^{\infty} u_i^2\Big) + \mathcal{O}_{u,0}\big(\|u\|_{\ell^1}^3\big). \tag{2.24a}$$

For each $k \geq 1$, we have

$$\frac{du_k}{dt} = \sigma_k(\chi_0)\, u_k - \frac{bA''}{2}\, \mathcal{S}_k^{(2)}(u) + \chi_0\, \mathcal{G}_k(u,v) - \frac{bA'''}{4}\, \mathcal{S}_k^{(3)}(u) + \mathcal{O}_{u,k}\big(\|u\|_{\ell^1}^4\big), \tag{2.24b}$$

where

$$\mathcal{S}_k^{(2)}(u) := \sum_{i-j=k} u_i u_j + \sum_{i-j=-k} u_i u_j + \sum_{i+j=k} u_i u_j, \tag{2.25}$$

$$\mathcal{S}_k^{(3)}(u) := \sum_{i+j+\ell=k} u_i u_j u_\ell + \sum_{i+j-\ell=k} u_i u_j u_\ell + \sum_{i+j-\ell=-k} u_i u_j u_\ell + \sum_{i-j+\ell=k} u_i u_j u_\ell + \sum_{i-j+\ell=-k} u_i u_j u_\ell + \sum_{i-j-\ell=k} u_i u_j u_\ell + \sum_{i-j-\ell=-k} u_i u_j u_\ell, \tag{2.26}$$

and

$$\mathcal{G}_k(u,v) := \frac{2}{L\chi_0} \sum_{\ell=1}^{4} \mathcal{F}_k^{(\ell)}(u,v) - \lambda_k B_0 C_k^{(1)} u_k = \mathcal{O}_{u,k}(\|u\|_{\ell^1}^2). \tag{2.27}$$

Step 4: In this step, we derive the ODE satisfied by u_{n_0}.

First, note that, on the center manifold $\mathcal{M}$ (see (2.16)), we have

$$\tilde{h}(u_{n_0}, \tilde{\chi}) = \big(h_0(u_{n_0}, \tilde{\chi}), \cdots, h_{n_0-1}(u_{n_0}, \tilde{\chi}), h_{n_0+1}(u_{n_0}, \tilde{\chi}), \cdots\big),$$

where

$$h_n(u_{n_0},\tilde{\chi}) = a_{n,1}\,u_{n_0}^2 + a_{n,2}\,\tilde{\chi}\,u_{n_0} + \mathcal{O}\big(|\tilde{\chi}|^2|u_{n_0}| + |\tilde{\chi}|u_{n_0}^2 + |u_{n_0}|^3\big), \quad n \neq n_0,$$

and

$$\sum_{n\neq n_0} |h_n(u_{n_0},\tilde{\chi})| < \infty \quad \forall |u_{n_0}| \ll 1 \quad \text{and} \quad |\tilde{\chi}| \ll 1.$$

Hence, the manifold

$$\widetilde{\mathcal{M}} = \Big\{ (U,\tilde{\chi}) = \big((u_n)_{n\geq 0},\tilde{\chi}\big) \big| u_{n_0} = o(1),\ \tilde{\chi} = o(1),\ u_n = h_n\big(u_{n_0},\tilde{\chi}\big) \text{ for } n \neq n_0 \Big\}$$

is locally invariant under (2.24a)+(2.24b)+(2.15) and contains all sufficiently small equilibria. For each fixed $\widetilde{\chi}$, the restriction of (2.24a)+(2.24b) to $\widetilde{\mathcal{M}}$ is the following scalar ODE:

$$\frac{du_{n_0}}{dt} = \big(\partial_{\chi_0}\sigma_{n_0}(\chi^*)\big)\tilde{\chi}\,u_{n_0} - \frac{bA''}{2}\,\mathcal{S}_{n_0}^{(2)}(u) + (\chi^* + \tilde{\chi})\,\mathcal{G}_{n_0}(u,v) - \frac{bA'''}{4}\,\mathcal{S}_{n_0}^{(3)}(u) + \mathcal{O}\big(\|u\|_{\ell^1}^4\big), \quad (2.28)$$

where $u = \{u_n\}$ with $u_n = h_n(u_{n_0},\tilde{\chi})$ for $n \neq n_0$.

In the following, we find the coefficients of the leading terms in (2.28). Since $h_k(0,\tilde{\chi}) = 0$, the coefficient of $\tilde{\chi}^2$ is zero. Hence, $u_k = h_k(u_{n_0},\tilde{\chi}) = a_{k,1}u_{n_0}^2 + a_{k,2}\tilde{\chi}u_{n_0} + \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big)$ for $k \geq 1$, $k \neq n_0$. We then obtain from (2.25), (2.26), and (2.27)

$$\mathcal{S}_k^{(2)}(u) = \begin{cases} u_{n_0}^2 + \mathcal{O}\Big(|\tilde{\chi}|\,|u_{n_0}|^2 + |u_{n_0}|^3\Big), & k = 2n_0, \\ \mathcal{O}\Big(|\tilde{\chi}|\,|u_{n_0}|^2 + |u_{n_0}|^3\Big), & k \neq 2n_0, \end{cases} \quad (2.29)$$

$$\mathcal{S}_k^{(3)}(u) = \mathcal{O}\Big(|\tilde{\chi}|^2|u_{n_0}| + |\tilde{\chi}|u_{n_0}^2 + |u_{n_0}|^3\Big), \quad (2.30)$$

and

$$\mathcal{G}_k(u,v) = \Gamma_k\,u_{n_0}^2 + \mathcal{O}\Big(|\tilde{\chi}|u_{n_0}^2 + |u_{n_0}|^3\Big), \quad (2.31)$$

where $\Gamma_k = 0$ for $k \neq 2n_0$, and

$$\Gamma_{2n_0} = \Big(\frac{n_0\pi}{L}\Big)^2\Big[4B_0\,C_{2n_0}^{(2)} + B_u\,C_{n_0}^{(1)} + B_v\,\big(C_{n_0}^{(1)}\big)^2\Big]. \quad (2.32)$$

Here B_0, B_u, and B_v are defined in (2.8), and $C_{n_0}^{(1)}$ and $C_{2n_0}^{(2)}$ are defined in (2.20).

We compute the coefficients of the leading terms in h_k. For $k = 0$, differentiating $u_0 = h_0(u_{n_0},\tilde{\chi})$ with respect to time and using (2.24b) with $k = n_0$, by (2.29), (2.30), and (2.31), we have that, for $|\tilde{\chi}| + |u_{n_0}|$ sufficiently small,

$$\frac{du_0}{dt} = \Big(2a_{0,1}u_{n_0} + a_{0,2}\tilde{\chi}\Big)\frac{du_{n_0}}{dt} + \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big) = \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big).$$

On the other hand, substituting $u_n = h_n(u_{n_0},\tilde{\chi})$ on the right-hand side of (2.24a) gives

$$\frac{du_0}{dt} = -\,a\alpha\,\Big(a_{0,1}u_{n_0}^2 + a_{0,2}\tilde{\chi}u_{n_0}\Big) - \frac{bA''}{2}u_{n_0}^2 + \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big).$$

By comparing the above two ODEs, we see that

$$a_{0,1} = -\frac{bA''}{2a\alpha} = -\frac{1+\alpha}{4u^*}, \qquad a_{0,2} = 0. \quad (2.33)$$

We can carry out the same procedure for u_k with $k \geq 1$ and $k \neq n_0$. Differentiating $u_k = h_k(u_{n_0}, \tilde{\chi})$ and using (2.24b) yields

$$\frac{du_k}{dt} = \Big(2a_{k,1}u_{n_0} + a_{k,2}\tilde{\chi}\Big)\frac{du_{n_0}}{dt} + \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big) = \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big). \tag{2.34}$$

Substitute $u_n = h_n(u_{n_0}, \tilde{\chi})$ into (2.24b). Using $u_k = a_{k,1}u_{n_0}^2 + a_{k,2}\tilde{\chi}u_{n_0} + \mathcal{O}(|\tilde{\chi}|^3 + |u_{n_0}|^3)$, we obtain

$$\frac{du_k}{dt} = \sigma_k(\chi^*)\Big(a_{k,1}u_{n_0}^2 + a_{k,2}\tilde{\chi}u_{n_0}\Big) - \frac{bA''}{2}\,\mathcal{S}_k^{(2)}(u) + \chi^*\,\mathcal{G}_k(u,v) + \mathcal{O}\Big(|\tilde{\chi}|^3 + |u_{n_0}|^3\Big). \tag{2.35}$$

By (2.29), (2.30), and (2.31) again, we have

$$a_{k,2} = 0 \quad \text{for all } k \neq n_0 \text{ with } k \geq 1, \tag{2.36a}$$

and

$$a_{k,1} = \begin{cases} \dfrac{\frac{bA''}{2} - \chi^*\Gamma_{2n_0}}{\sigma_{2n_0}(\chi^*)}, & k = 2n_0, \\ 0, & k \neq n_0,\, 2n_0 \text{ with } k \geq 1. \end{cases} \tag{2.36b}$$

By (2.33) and (2.36a), $a_{k,2} = 0$ for every $k \neq n_0$; hence smoothness of the center-manifold map and $h_k(0, \widetilde{\chi}) = 0$ give

$$h_k(u_{n_0}, \widetilde{\chi}) = a_{k,1}u_{n_0}^2 + \mathcal{O}\left(|\widetilde{\chi}|^2|u_{n_0}| + |\widetilde{\chi}||u_{n_0}|^2 + |u_{n_0}|^3\right). \tag{2.37}$$

We then have

$$\mathcal{S}_{n_0}^{(2)}(u) = \left(4a_{0,1} + 2a_{2n_0,1}\right)u_{n_0}^3 + \mathcal{O}\Big(|u_{n_0}|^2|\tilde{\chi}|^2 + |u_{n_0}|^3|\tilde{\chi}| + |u_{n_0}|^4\Big), \tag{2.38}$$

$$\mathcal{S}_{n_0}^{(3)}(u) = 3\,u_{n_0}^3 + \mathcal{O}\Big(|u_{n_0}|^2|\tilde{\chi}|^2 + |u_{n_0}|^3|\tilde{\chi}| + |u_{n_0}|^4\Big), \tag{2.39}$$

and

$$\mathcal{G}_{n_0}(u,v) = \Gamma_{n_0}^{(3)}\,u_{n_0}^3 + \mathcal{O}\Big(|u_{n_0}|^2|\tilde{\chi}|^2 + |u_{n_0}|^3|\tilde{\chi}| + |u_{n_0}|^4\Big), \tag{2.40}$$

where the $\Gamma_{n_0}^{(3)}$ coefficient is

$$\Gamma_{n_0}^{(3)} = \Big(\frac{n_0\pi}{L}\Big)^2\Big[B_0H + \Big(B_u + B_vF\Big)G + F\Big(R - \frac{S}{2}\Big)\Big], \tag{2.41}$$

B_0, B_u, B_v are defined in (2.8),

$$F \coloneqq C_{n_0}^{(1)}, \quad G \coloneqq C_{2n_0}^{(1)}a_{2n_0,1} + C_{2n_0}^{(2)}, \quad H \coloneqq 2C_{n_0}^{(2)}\Big(2a_{0,1} + a_{2n_0,1}\Big) + 3C_{n_0}^{(3)}, \tag{2.42}$$

and

$$R \coloneqq B_ua_{0,1} + B_v\big(C_0^{(1)}a_{0,1} + C_0^{(2)}\big) + \frac{Q}{2},\ S \coloneqq B_ua_{2n_0,1} + B_vG + \frac{Q}{2},\ Q \coloneqq B_{uu} + B_{uv}F + B_{vv}F^2.$$

Substituting the expansion (2.37) into the quadratic and cubic terms of (2.28), using (2.38), (2.39), (2.40), and the fact that $h_n(0, \tilde{\chi}) = 0$ for $|\tilde{\chi}| \ll 1$, we must have

$$\frac{du_{n_0}}{dt} = \alpha_{n_0}\,\tilde{\chi}\,u_{n_0} - \beta_{n_0}\,u_{n_0}^3 + \mathcal{O}\Big(|u_{n_0}|^2|\tilde{\chi}|^2 + |u_{n_0}|^3|\tilde{\chi}| + |u_{n_0}|^4\Big), \tag{2.43}$$

where

$$\alpha_{n_0} = \partial_{\chi_0}\sigma_{n_0}(\chi^*) = \nu\gamma\,\frac{(u^*)^{m+\gamma-1}}{(1+v^*)^{\beta}}\,\frac{\lambda_{n_0}}{\mu+\lambda_{n_0}} > 0, \tag{2.44}$$

$$\beta_{n_0} = \frac{(1+\alpha)\alpha\, b\,(u^*)^{\alpha-1}}{4}\big(4a_{0,1}+2a_{2n_0,1}\big) - \chi^*\,\Gamma^{(3)}_{n_0} + \frac{(1+\alpha)\alpha(\alpha-1)b\,(u^*)^{\alpha-2}}{8}. \tag{2.45}$$

Step 5: In this step, we conclude the bifurcation analysis. Writing $z = u_{n_0}$, $\tau = \widetilde{\chi}$, and R for the remainder in (2.43), Taylor expansion of the C^4 reduced vector field, with its exact linear term and the cancellations in Step 4, gives

$$|R_z(\tau, z)| \le C\left(|z||\tau|^2 + |z|^2|\tau| + |z|^3\right).$$

This estimate justifies local uniqueness of each nonzero branch and the stability conclusion below.

By (2.43), when $\beta_{n_0} \neq 0$ and $|\tilde{\chi}|, |u_{n_0}| \ll 1$, (2.43) has two branches $u^{\pm}_{n_0}(\tilde{\chi})$ of equilibrium solutions on the side $\beta_{n_0}\widetilde{\chi} > 0$, with

$$u^{\pm}_{n_0}(\widetilde{\chi}) = \pm\sqrt{\frac{\alpha_{n_0}}{\beta_{n_0}}\widetilde{\chi}} + \mathcal{O}(|\widetilde{\chi}|), \tag{2.46}$$

which gives two branches $u^{\pm}$ of equilibrium solutions of (1.3) on the side $\beta_{n_0}(\chi_0 - \chi^*) > 0$ given in Theorem 1.1. Moreover, the derivative of the reduced field on either branch is $-2\alpha_{n_0}\widetilde{\chi} + o(|\widetilde{\chi}|)$; together with the stable complementary spectrum, this implies that when $\beta_{n_0} > 0$, the bifurcation occurs when $\chi_0 > \chi^*$ and the bifurcation solutions $u^{\pm}$ are linearly asymptotically stable, which is referred to as super-critical pitchfork bifurcation, and when $\beta_{n_0} < 0$, the bifurcation occurs when $\chi_0 < \chi^*$ and solutions $u^{\pm}$ are linearly unstable, which is referred to as sub-critical pitchfork bifurcation. □

2.2 Local pitchfork bifurcation and proof of Theorem 1.1 for the minimal model

In this subsection, we prove Theorem 1.1 for the minimal model.

Proof of Theorem 1.1 for the minimal model. Throughout this proof, assume $a = b = 0$ and fix the positive constant equilibrium (u^*, v^*), where $v^* = \frac{\nu}{\mu}(u^*)^{\gamma}$. As in the non-minimal case, set

$$u(t,x) = u^* + \widetilde{u}(t,x), \qquad v(t,x) = v^* + \widetilde{v}(t,x).$$

Then $(\widetilde{u}, \widetilde{v})$ satisfies (2.6)–(2.7) with $a = b = 0$, and $\widetilde{v} = \mathcal{K}(\widetilde{u})$, where $\mathcal{K}$ is defined by (2.12). Consequently, $\widetilde{u}$ satisfies (2.13) with $b = 0$.

The proof follows the same arguments as in the non-minimal model. We give the main differences. We divide the proof into four steps.

Step 1′: We identify the invariant fixed-mass phase space and construct the center manifold.

Recall the fractional power space X_θ introduced earlier in Section 2.1, where $3/4 < \theta < 1$. Define

$$\hat{X}_\theta := \Big\{w \in X_\theta : \int_0^L w(x)\,dx = 0\Big\}.$$

Then

$$X_\theta = \hat{X}_\theta \oplus \operatorname{span}\{1\}.$$

It is easy to see that $\hat{X}_\theta$ is invariant under (2.6)–(2.7) with $a = b = 0$.

Set

$$\hat{X}_\theta = \mathrm{span}\{\phi_{n_0}\} \oplus \hat{X}^s_\theta.$$

Since n_0 is the unique bifurcation mode, the linearization at $(\widetilde{u}, \chi_0) = (0, \chi^*)$, restricted to $\hat{X}_\theta$, has the simple eigenvalue $\sigma_{n_0}(\chi^*) = 0$, while $\sigma_k(\chi^*) < 0$ for every $k \geq 1$, $k \neq n_0$.

As in Step 1 of the proof for the non-minimal model, there is a locally invariant manifold

$$\mathcal{M}_0 = \left\{ \left(\hat{h}(u_{n_0}, \widetilde{\chi}) + u_{n_0}\phi_{n_0}, \widetilde{\chi} \right) : |u_{n_0}| \ll 1,\ |\widetilde{\chi}| \ll 1 \right\}$$

of (2.13) with $b = 0$, where $\hat{h} : \mathrm{span}\{\phi_{n_0}\} \times \mathbb{R} \longrightarrow \hat{X}^s_\theta$ is a C^4 map satisfying $\hat{h}(0,0) = 0$, $D\hat{h}(0,0) = 0$, $\hat{h}(0, \widetilde{\chi}) = 0$ for all $|\widetilde{\chi}| \ll 1$.

Step 2′: We derive the Fourier-mode equations on the fixed-mass space.

For a solution on $\mathcal{M}_0$, write

$$\widetilde{u}(t,x) = \sum_{k=1}^{\infty} u_k(t) \cos\left(\tfrac{k\pi x}{L}\right), \qquad \widetilde{v}(t,x) = \sum_{k=0}^{\infty} v_k(t) \cos\left(\tfrac{k\pi x}{L}\right).$$

The sum for $\widetilde{u}$ begins at $k = 1$ because mass conservation gives $u_0(t) = 0$. In contrast, v_0 need not vanish when $\gamma \neq 1$. Setting $u_0 = 0$ in (2.18) gives

$$v_0 = C_0^{(2)} \sum_{j=1}^{\infty} u_j^2 + \frac{\nu\gamma(\gamma-1)(\gamma-2)(u^*)^{\gamma-3}}{8\mu} \sum_{i,j\geq 1} u_i u_j u_{i+j} + \mathcal{O}_{v,0}\left(\|u\|_{\ell^1}^4\right).$$

For $k \geq 1$, the coefficients v_k are given by (2.19) with $u_0 = 0$.

Projecting the u-equation onto the nonconstant cosine modes and setting $b = 0$ in (2.24b) yields

$$\frac{du_k}{dt} = \sigma_k(\chi_0)u_k + \chi_0 \mathcal{G}_k(u,v) + \mathcal{O}_{u,k}\left(\|u\|_{\ell^1}^4\right), \qquad k \geq 1.$$

Therefore, near $\chi_0 = \chi^*$,

$$\begin{cases} \dfrac{du_{n_0}}{dt} = \left(\partial_{\chi_0}\sigma_{n_0}(\chi^*)\right)\widetilde{\chi}u_{n_0} + (\chi^* + \widetilde{\chi})\mathcal{G}_{n_0}(u,v) + \mathcal{O}_{u,n_0}\left(\|u\|_{\ell^1}^4\right), \\ \dfrac{du_k}{dt} = \left(\sigma_k(\chi^*) + (\partial_{\chi_0}\sigma_k)(\chi^*)\widetilde{\chi}\right)u_k + (\chi^* + \widetilde{\chi})\mathcal{G}_k(u,v) + \mathcal{O}_{u,k}\left(\|u\|_{\ell^1}^4\right), \quad k \geq 1,\ k \neq n_0, \\ \dfrac{d\widetilde{\chi}}{dt} = 0. \end{cases} \tag{2.47}$$

Step 3′: We compute the reduced equation on $\mathcal{M}_0$ through cubic order.

By the same arguments as in the non-minimal case, we get the scalar reduced equation

$$\frac{du_{n_0}}{dt} = \alpha_{n_0}\widetilde{\chi}u_{n_0} - \beta_{n_0}u_{n_0}^3 + \mathcal{O}\left(|u_{n_0}|^2|\widetilde{\chi}|^2 + |u_{n_0}|^3|\widetilde{\chi}| + |u_{n_0}|^4\right), \tag{2.48}$$

with

$$\beta_{n_0} = -\chi^*\Gamma^{(3)}_{n_0}, \tag{2.49}$$

and α_{n_0} as in (1.13), where we set $a = b = 0$, and $a_{0,1} = 0$ due to the fixed-mass constraint.

Step 4′: We use the same argument as in Step 5 for the non-minimal model, now in the fixed-mass space. The same Taylor argument gives the derivative remainder estimate used there, with the constant mode excluded. This proves Theorem 1.1 in the minimal case. □

3 Effects of β and L on the bifurcation and proof of Theorem 1.2

In this section, we study the effects of the nonlinear sensitivity exponent β and the domain size L on the bifurcation of positive constant solutions of (1.3) and prove Theorem 1.2.

3.1 Effects of β and proof of Theorem 1.2(1)

In this subsection, we explore the changes of bifurcation direction as β changes, but all the other parameters are fixed, and prove Theorem 1.2(1).

Proof of Theorem 1.2(1). We first prove (1.14). Recall

$$\beta_{n_0} = \frac{(1+\alpha)\alpha\, b\,(u^*)^{\alpha-1}}{4}\big(4a_{0,1} + 2a_{2n_0,1}\big) - \chi^*\,\Gamma^{(3)}_{n_0} + \frac{(1+\alpha)\alpha(\alpha-1)b\,(u^*)^{\alpha-2}}{8}$$

(see (2.45)). Throughout this proof, we assume that all the parameters are fixed, except β. We first find the dependence of $a_{2n_0,1}$ and $\chi^*\Gamma^{(3)}_{n_0}$ on β.

First, for simplicity, define

$$p(\beta) := a_{2n_0,1}(\beta) = \frac{\frac{bA''}{2} - \chi^*\Gamma_{2n_0}}{\sigma_{2n_0}(\chi^*)}, \qquad q(\beta) := \chi^*(u^*,\beta)\Gamma^{(3)}_{n_0}(\beta).$$

Using this notation, formula (2.45) becomes

$$\beta_{n_0}(\beta) = \frac{bA''}{2}\,(4a_{0,1} + 2p(\beta)) - q(\beta) + \frac{3bA'''}{4}. \tag{3.1}$$

Set

$$\Pi := \frac{\chi^*(u^*,\beta)}{(1+v^*)^\beta}\lambda_{n_0} \overset{\text{by (1.8)}}{=} \frac{(\lambda_{n_0}+a\alpha)(\mu+\lambda_{n_0})}{\nu\gamma\,(u^*)^{m+\gamma-1}}. \tag{3.2}$$

Hence, Π is positive and independent of β. It follows from (2.32) that

$$\chi^*\Gamma_{2n_0} = \Pi\Big(4(u^*)^m C^{(2)}_{2n_0} + m(u^*)^{m-1}C^{(1)}_{n_0} - \beta\frac{(u^*)^m\big(C^{(1)}_{n_0}\big)^2}{1+v^*}\Big).$$

Hence, $\chi^*\Gamma_{2n_0}$ is affine in β. Since

$$\sigma_{2n_0}(\chi^*) = -4\lambda_{n_0} + \Pi\frac{4\nu\gamma(u^*)^{m+\gamma-1}}{\mu+4\lambda_{n_0}} - a\alpha,$$

the function p is also affine in β. This together with (2.41) and (3.2) shows that q is a polynomial of degree at most two.

Because p is affine and q is quadratic, the right-hand side of (3.1) is a quadratic polynomial in β. Its Taylor expansion at $\beta = 0$ is exact and gives

$$\beta_{n_0}(\beta) = A\beta^2 + B\beta + C,$$

where

$$A = -\frac{1}{2}q''(0), \quad B = bA''p'(0) - q'(0), \quad C = \frac{bA''}{2}\,(4a_{0,1} + 2p(0)) - q(0) + \frac{3bA'''}{4}. \tag{3.3}$$

A direct differentiation of (2.41) twice with respect to β gives

$$A = \frac{\left(C_{n_0}^{(1)}\right)^3 (u^*)^m \Pi \left(4C_{2n_0}^{(1)}(u^*)^m \Pi - \sigma_{2n_0}(\chi^*)\right)}{8\sigma_{2n_0}(\chi^*)(1+v^*)^2}. \tag{3.4}$$

We now determine its sign. By the definitions of the coefficients,

$$C_{n_0}^{(1)} > 0, \qquad C_{2n_0}^{(1)} > 0, \qquad u^* > 0, \qquad \Pi > 0.$$

In the case that $a, b > 0$, since n_0 is the unique bifurcation mode and $2n_0 \neq n_0$, we have

$$\sigma_{2n_0}(\chi^*) < 0, \quad \text{and then } 4C_{2n_0}^{(1)}(u^*)^m \Pi - \sigma_{2n_0}(\chi^*) > 0. \tag{3.5}$$

Therefore, the numerator in (3.4) is positive. The denominator is negative because $8\sigma_{2n_0}(\chi^*)(1+v^*)^2 < 0$. Consequently, $A < 0$. In particular, A cannot vanish under the unique bifurcation mode assumption.

In the minimal model, we impose the fixed-mean constraint, so $a_{0,1} = 0$. We retain the notation $p(\beta)$, $q(\beta)$, and Π, with the coefficients evaluated for $a = b = 0$ and $a_{0,1} = 0$. In this case, we have $\beta_{n_0}(\beta) = -q(\beta) = A\beta^2 + B\beta + C$, where

$$A = -\frac{1}{2}q''(0), \qquad B = -q'(0), \qquad C = -q(0).$$

The calculation of the quadratic coefficient is unchanged and gives the same expression as (3.4), where $\sigma_{2n_0}(\chi^*)$ is now evaluated at $a = 0$. Since $n_0 = 1$ is the unique bifurcation mode on the fixed-mean space, $\sigma_{2n_0}(\chi^*) < 0$, because $2n_0 \notin \{0, n_0\}$. All the other factors in the numerator are positive, and (3.5) holds. Hence, the numerator is positive and the denominator is negative. It follows that $A < 0$ in the minimal model as well. Thus, in both the non-minimal and minimal cases, $\beta_{n_0}(\beta)$ is a quadratic polynomial with a strictly negative leading coefficient.

In the following, we show that one of the cases (i)-(iv) in Theorem 1.2(1) occurs. Let $D \coloneqq B^2 - 4AC$ and β_- and β_+ be the roots of $\beta_{n_0}(\beta)$. We then have

(i) If $C > 0$, there is exactly one positive root β_+, with $\beta_{n_0} > 0$ on $[0, \beta_+)$ and $\beta_{n_0} < 0$ on (β_+, ∞).

(ii) If $C \leq 0$, $B > 0$, and $D > 0$, there are two nonnegative roots $0 \leq \beta_- < \beta_+$, with $\beta_{n_0} > 0$ precisely between them. The equality $\beta_- = 0$ occurs when $C = 0$.

(iii) If either $C < 0$, $B > 0$, and $D = 0$, or $C = 0$ and $B \leq 0$, there is one nonnegative root β_0 at which $\beta_{n_0}(\beta_0) = 0$, and $\beta_{n_0}(\beta) < 0$ for every other $\beta \geq 0$.

(iv) In all remaining cases with $C < 0$, one has $\beta_{n_0}(\beta) < 0$ for every $\beta \geq 0$.

This concludes the proof of Theorem 1.2(1). □

Table 3.1 illustrates several of the sign patterns in Theorem 1.2(1); it does not include a zero of the cubic coefficient. For the non-minimal model (left panel), $\Omega = (0, 1)$ and $n_0 = 1$. The parameter vector $(a, b, \alpha, m, \gamma, \nu, \mu, L)$ equals $(1, 1, 1, 1, 1, 1, 1, 1)$ in the first block, $(1, 1, 3, 1, 2, 10, 100, 1)$ in the second block, and $(1, 1, 5, 1, 1, 1, 1, 1)$ in the third block. For the minimal model (right panel), $a = b = 0$. The parameter vector $(u^*, m, \gamma, \nu, \mu, L)$ equals $(1, 1, 1, 1, 1, 4)$ in the first block, $(3, 1, 2, 1, 1, 4)$ in the second block, and $(1, \frac{1}{3}, \frac{1}{2}, 1, 1, 4)$ in the third block.

Table 3.1: Bifurcation directions for the non-minimal model and the minimal model.

Non-minimal model				
(m,γ,n_0)	β	$\chi^*(u^*)$	β_1	Type
$(1,1,1)$	0	11.9709	+5.7006	super-critical
	2	47.8837	+0.4202	super-critical
	3	95.7674	−2.2322	sub-critical
$(1,2,1)$	1	7.8796	−17.4850	sub-critical
	5	11.5366	+8.0931	super-critical
	10	18.5798	−3.6136	sub-critical
$(1,1,1)$	0	16.3762	−2.9280	sub-critical
	1	32.7524	−14.3380	sub-critical
	5	524.0387	−60.1009	sub-critical

Minimal model				
(u^*,m,γ,n_0)	β	$\chi^*(u^*)$	β_{n_0}	Type
$(1,1,1,1)$	0	1.6169	+0.0097	super-critical
	4	25.8696	+0.0212	super-critical
	5	51.7392	−0.0450	sub-critical
$(3,1,2,1)$	0	0.0898	−0.0299	sub-critical
	1	0.8983	+0.0228	super-critical
	4	898.2502	−0.0577	sub-critical
$(1,\frac{1}{3},\frac{1}{2},1)$	0	3.2337	−0.0211	sub-critical
	1	6.4674	−0.0512	sub-critical
	2	12.9348	−0.0881	sub-critical

3.2 Effects of domain size L and proof of Theorem 1.2(2)

In this subsection, we explore the changes of bifurcation direction and bifurcation modes as the domain size L changes, but all the other parameters are fixed, and prove Theorem 1.2(2).

Proof of Theorem 1.2(2). Recall that

$$\beta_{n_0}=\frac{(1+\alpha)\alpha\, b\,(u^*)^{\alpha-1}}{4}\big(4a_{0,1}+2a_{2n_0,1}\big)-\chi^*\,\Gamma^{(3)}_{n_0}+\frac{(1+\alpha)\alpha(\alpha-1)b\,(u^*)^{\alpha-2}}{8}$$

(see (2.45)). Note that β_{n_0} depends on L since $a_{2n_0,1}$ and $\chi^*\Gamma^{(3)}_{n_0}$ depend on $\lambda_{n_0}=\frac{n_0^2\pi^2}{L^2}$. Put

$$K=\frac{\nu\gamma(u^*)^{m+\gamma-1}}{(1+v^*)^{\beta}}.$$

We first establish the rational representation. In the non-minimal model, it is not difficult to see that

$$\chi^*=\frac{(\lambda+a\alpha)(\mu+\lambda)}{K\lambda},\qquad a_{2n_0,1}=\frac{N_3(\lambda)}{3K(\mu+\lambda)(a\alpha\mu-4\lambda^2)},$$

where N_3 is a polynomial of degree at most three, and

$$\chi^*\Gamma^{(3)}_{n_0}=\frac{(\lambda+a\alpha)N_4(\lambda)}{12K^2(\mu+\lambda)^2(a\alpha\mu-4\lambda^2)},$$

where N_4 is a polynomial of degree at most four. Substitution into the formula for β_{n_0} gives

$$\beta_{n_0}=\frac{P(\lambda)}{12K^2(\mu+\lambda)^2(4\lambda^2-a\alpha\mu)},$$

where $P(\lambda)$ is a polynomial of degree at most 5 given by

$$P(\lambda)=-12K^2\big(2bA''a_{0,1}+3bA'''/4\big)(\mu+\lambda)^2(a\alpha\mu-4\lambda^2)-4KbA''(\mu+\lambda)N_3+(\lambda+a\alpha)N_4.$$

In the minimal model, $n_0=1$, $a_{0,1}=0$, and

$$\chi^*=\frac{\mu+\lambda}{K},\qquad \sigma_2(\chi^*)=-\frac{12\lambda^2}{\mu+4\lambda}.$$

The same substitution and simplification yield

$$\beta_1 = \frac{P(\lambda)}{24(u^*)^2(\mu+\lambda)^2}, \qquad \deg P \le 3.$$

Next, we consider the sign changes of $P(\lambda_{n_0})$. Recall that, for every $k \ge 1$,

$$n_0 = k \text{ is the unique minimizing mode exactly for } \quad L \in (L_{k-1}, L_k)\,, \tag{3.6}$$

while at $L = L_k$ the infimum is attained at the two modes k and $k+1$ and at no others.

In the non-minimal model, $n_0 = k$ is the unique minimizing mode when $L \in (L_{k-1}, L_k)$. On these intervals,

$$\frac{1}{2}\sqrt{a\alpha\mu} < \lambda_{n_0} = \lambda_1 < \infty \qquad (k=1),$$

and

$$\frac{k}{k+1}\sqrt{a\alpha\mu} < \lambda_{n_0} = \lambda_k < \frac{k}{k-1}\sqrt{a\alpha\mu} \qquad (k \ge 2).$$

In the minimal model, $n_0 = 1$ for every $L > 0$, and λ_1 ranges over $(0, \infty)$. In either case, λ_{n_0} decreases strictly on each interval with fixed mode. Since the denominator is positive, a nonzero P changes sign at its odd-multiplicity zeros. This proves part (i), including the degree bounds.

Finally, in the non-minimal model,

$$\lambda_{n_0(L)} \longrightarrow \lambda_\infty := \sqrt{a\alpha\mu} \quad (L \to \infty)$$

along admissible lengths. The rational expression is continuous at λ_∞, so its limit defines β_∞. In the minimal model, $\lambda_1 \to 0$, and direct calculation gives

$$P(0) = 24(u^*)^2\mu^2\beta_\infty^{\min}.$$

Hence, $\beta_1 \to \beta_\infty^{\min}$.

In both models, polynomial division gives

$$\beta_{n_0}(\lambda) = c\lambda + \mathcal{O}(1) \quad (\lambda \to \infty),$$

with $c = c_{\min}$ in the minimal case. Since $n_0 = 1$ for all sufficiently small L,

$$\beta_{n_0} = c\frac{\pi^2}{L^2} + \mathcal{O}(1) \quad (L \to 0+).$$

The signs of β_∞ and c, when nonzero, prove parts (ii) and (iii), respectively. □

Example 3.1. Computation of β_{n_0} and switch of bifurcation regimes when $a = b = \mu = \nu = m = \alpha = \gamma = 1$

In this example, we discuss the switch of bifurcation regimes when $a = b = \mu = \nu = m = \alpha = \gamma = 1$. Note that, in this case, we have $u^* = v^* = 1$. Also, by direct calculation, we have

$$\begin{aligned}\beta_{n_0} &= \frac{P(\lambda_{n_0},\beta)}{Q(\lambda_{n_0},\beta)} \\ &= \frac{(224-96\beta)\lambda_{n_0}^4+(192-152\beta)\lambda_{n_0}^3+(8-104\beta-16\beta^2)\lambda_{n_0}^2+(96-28\beta-8\beta^2)\lambda_{n_0}+56+11\beta-\beta^2}{96(1+\lambda_{n_0})(4\lambda_{n_0}^2-1)}.\end{aligned}$$

Since $Q(\lambda_{n_0}, \beta) > 0$ for $\lambda_{n_0} > 1/2$, the sign of β_{n_0} is determined by $P(\lambda_{n_0}, \beta)$.

Writing $\lambda = \lambda_{n_0}$, we obtain

$$\beta_{n_0} = \Big(\frac{7}{12} - \frac{\beta}{4}\Big)\lambda + \mathcal{O}(1) \qquad (\lambda \to \infty). \tag{3.7}$$

Thus, the small-domain bifurcation is super-critical for $\beta < 7/3$ and sub-critical for $\beta > 7/3$. For large admissible domains, $\lambda_{n_0} \to 1$, giving

$$\beta_\infty = 1 - \frac{41}{64}\beta - \frac{25}{576}\beta^2. \tag{3.8}$$

Hence, the large-domain bifurcation is super-critical for $\beta < \beta_{\mathrm{ld}}$ and sub-critical for $\beta > \beta_{\mathrm{ld}}$, where

$$\beta_{\mathrm{ld}} = \frac{3\sqrt{21529} - 369}{50} \approx 1.423658.$$

β	Roots	β	Roots
1	None	2.1	7.370444
1.5	0.773411, 1.518665	7/3	None
1.9	0.530181, 3.826548	2.4	None
2	5.092791	3	None

Table 3.2: Roots of $P(\lambda, \beta)$ with $\lambda > 1/2$ for selected values of β, rounded to six decimal places.

For $\lambda > 1/2$, $P(\lambda, \beta)$ decreases strictly with β. Moreover, $P(\lambda, 1) > 0$ and $P(\lambda, 7/3) < 0$. Thus, the bifurcation is super-critical for $0 \le \beta \le 1$ and sub-critical for $\beta \ge 7/3$ at every admissible length.

For intermediate values of β, the roots in Table 3.2, together with the eigenvalue intervals in Theorem 1.2(2), give the following examples as L increases.

For instance, for $\beta = 1.5$ and $\beta = 1.9$, the positive roots are

β	λ_-	λ_+
1.5	0.773411	1.518665
1.9	0.530181	3.826548

(values rounded), with

$$\operatorname{sgn}\beta_{n_0} = \begin{cases} +, & \tfrac{1}{2} < \lambda < \lambda_- \text{ or } \lambda > \lambda_+, \\ -, & \lambda_- < \lambda < \lambda_+. \end{cases}$$

Since $\lambda_k = k^2\pi^2/L^2$ decreases as L increases on (L_{k-1}, L_k), the signs follow the sequences

β	$k = 1$	$k = 2$	$k \ge 3$
1.5	$+ \to - \to +$	$+ \to - \to +$	$\begin{cases} - \to +, & k = 3, \\ -, & k \ge 4, \end{cases}$
1.9	$+ \to - \to +$	$-$	$-$

where $+$ and $-$ denote super-critical and sub-critical bifurcation, respectively; the cubic coefficient vanishes at each transition.

4 Extension of the local bifurcation branches and proof of Theorem 1.3

In this section, we study the extension of the local bifurcation branches and prove Theorem 1.3.

We first collect some estimates for stationary solutions of (1.3) to be used in the proofs of Theorem 1.3 as well as Theorem 1.4. Throughout, $\alpha, \gamma, m, \mu, \nu > 0$, $\beta \geq 0$.

Lemma 4.1. *Assume $a, b > 0$. Let $u^* = \left(\frac{a}{b}\right)^{\frac{1}{\alpha}}$. Then the following hold.*

(1) For every $\alpha, \gamma, m > 0$, and every nonconstant positive stationary solution (u, v) of (1.3),

$$0 < \min_{[0,L]} u < u^* < \max_{[0,L]} u, \quad \text{and} \quad \int_0^L u\,dx < Lu^*, \quad \int_0^L u^{1+\alpha}\,dx < L\,(u^*)^{1+\alpha}. \tag{4.1}$$

(2) Assume $0 < \gamma \leq 1 + \alpha$ and $1 \leq m \leq \alpha + 2$. For every $\overline{X} > 0$, there exist $\theta \in (0, 1)$ and $C = C(\overline{X}, a, b, \alpha, \gamma, \mu, \nu, L, m, \beta) > 0$ such that every positive nonconstant stationary solution with $0 \leq \chi_0 \leq \overline{X}$ satisfies

$$\|u\|_{C^{2+\theta}([0,L])},\ \|v\|_{C^{2+\theta}([0,L])} \leq C. \tag{4.2}$$

Proof. (1) Note that $u_t = 0$. Then by the first equation in (1.3), we have

$$\int_0^L \left(au - bu^{1+\alpha}\right)dx = 0, \quad \text{hence} \quad a\int_0^L u\,dx = b\int_0^L u^{1+\alpha}\,dx. \tag{4.3}$$

Since u is positive and nonconstant, $u - u^*$ changes sign. Hence,

$$0 < \inf_{x\in[0,L]} u(x) < u^* < \sup_{x\in[0,L]} u(x).$$

By strict Hölder's inequality, it holds

$$\Big(\int_0^L u\,dx\Big)^{1+\alpha} < L^\alpha \int_0^L u^{1+\alpha}\,dx = L^\alpha (u^*)^\alpha \int_0^L u\,dx.$$

(4.1) then follows.

(2) Fix $\overline{X} > 0$ and assume $0 \leq \chi_0 \leq \overline{X}$. Let (u, v) be a positive nonconstant stationary solution of (1.3). First, note that

$$0 = u_{xx} + b_1(x)u_x + c_1(x), \quad 0 < x < L, \ \ u_x(0) = u_x(L) = 0, \tag{4.4}$$

and

$$0 = v_{xx} + b_2(x)v + c_2(x), \quad 0 < x < L, \ \ v_x(0) = v_x(L) = 0, \tag{4.5}$$

where $b_1(x) = -\chi_0 m \frac{u^{m-1}(x)v_x(x)}{(1+v)^\beta}$, $c_1(x) = -\chi_0 \frac{u^m}{(1+v)^\beta}(\mu v - \nu u^\gamma) + \chi_0 \beta u^m \frac{|v_x|^2}{(1+v)^{1+\beta}} + u(a - bu^\alpha)$, $b_2(x) = -\mu$, $c_2(x) = \nu u^\gamma(x)$. By a priori estimates for elliptic equations, it then suffices to prove that $b_1(x)$, $c_1(x)$, $b_2(x)$, and $c_2(x)$ are uniformly Hölder continuous on $[0, L]$. For this, it suffices to prove that u, u_x, v, v_x, and v_{xx} are uniformly bounded on $[0, L]$.

Next, we prove the uniform boundedness of $\|v\|_\infty$ and $\|v_x\|_\infty$. Note that for any $0 \leq \eta \leq 1 + \alpha$, Hölder's inequality and (4.1) yield

$$M_\eta := \int_0^L u^\eta\,dx \leq L\,(u^*)^\eta. \tag{4.6}$$

By the second equation in (1.3), we have

$$\int_0^L v(x)dx = \frac{\nu}{\mu}M_\gamma \le \frac{\nu}{\mu}L(u^*)^\gamma, \quad \text{and} \quad v_x(x) = \mu\int_0^x v(x)dx - \nu\int_0^x u^\gamma(x)dx.$$

It then follows that

$$\|v_x\|_\infty \le \mu\int_0^L v\,dx + \nu\int_0^L u^\gamma\,dx = 2\nu\int_0^L u^\gamma\,dx \le 2\nu L\,(u^*)^\gamma\,, \tag{4.7}$$

and there is $x_0 \in [0, L]$ such that $v(x_0) \le \frac{\nu}{\mu}(u^*)^\gamma$, which implies that

$$\|v\|_\infty \le |v(x_0)| + \sup_{x\in[0,L]}\Big|\int_{x_0}^x v_y(y)dy\Big| \le \frac{\nu}{\mu}(u^*)^\gamma + 2\nu L^2\,(u^*)^\gamma\,. \tag{4.8}$$

Now, we prove the uniform boundedness of $\|u\|_\infty$ and $\|u_x\|_\infty$. Notice that

$$u_x = \chi_0\frac{u^m}{(1+v)^\beta}v_x - F(x), \qquad F(x) := \int_0^x \left(au - bu^{1+\alpha}\right)\,dy, \tag{4.9}$$

and

$$\|F\|_\infty \le a\int_0^L u\,dx + b\int_0^L u^{1+\alpha}\,dx < 2aLu^*. \tag{4.10}$$

Choose $x_0 \in [0, L]$ such that $u(x_0) < u^*$. We then have

$$u(x) = u(x_0)e^{\chi_0\int_{x_0}^x \frac{u^{m-1}(y)}{(1+v(y))^\beta}v_y(y)dy} - \int_{x_0}^x e^{\chi_0\int_z^x \frac{u^{m-1}(y)}{(1+v(y))^\beta}v_y(y)dy}F(z)dz \quad \forall\, x \in [0, L].$$

This together with (4.6) with $\eta = m - 1(\le \alpha + 1)$, (4.7), (4.9) and (4.10) implies that

$$\|u\|_\infty \le C \quad \text{and} \quad \|u_x\|_\infty \le C \tag{4.11}$$

for some $C = C(\overline{X}, a, b, \alpha, \gamma, \mu, \nu, L, m, \beta) > 0$.

Finally, by the second equation in (1.3) again, we have

$$\|v_{xx}\|_\infty \le \mu\|v\|_\infty + \nu\|u^\gamma\|_\infty. \tag{4.12}$$

By $m \ge 1$ and the uniform bounds of $\|u\|_\infty$, $\|u_x\|_\infty$, $\|v\|_\infty$, $\|v_x\|_\infty$, and $\|v_{xx}\|_\infty$, the coefficients b_1, c_1, b_2, and c_2 in (4.4) and (4.5) are in $C^\theta([0, L])$ with $0 < \theta < \min\{1, \gamma\}$ when $m = 1$ and $0 < \theta < \min\{1, \gamma, m - 1\}$ when $m > 1$. Then, by a priori estimates for elliptic equations, there is $C = C(\overline{X}, a, b, \alpha, \gamma, \mu, \nu, L, m, \beta) > 0$ such that (4.2) holds. □

Similarly, we have the following analogue of Lemma 4.1 for the minimal model.

Lemma 4.2. *Assume that $a = b = 0$, and $1 \le m \le 2$, $0 < \gamma \le 1$. Let (u^*, v^*) be a positive constant stationary solution of (1.3). Then the following hold.*

(1) Let (u, v) be a positive nonconstant stationary solution of (1.3) with $\int_0^L u(x)dx = Lu^*$. It holds that $0 < \min_{[0,L]} u < u^* < \max_{[0,L]} u$ and $\int_0^L u^2\,dx > L(u^*)^2$.

(2) For every $\bar{X} > 0$, there are $\theta \in (0, 1)$ and $C = C(\overline{X}, u^*, \mu, \nu, \beta, \gamma, L, m) > 0$ such that every positive nonconstant stationary solution with $\int_0^L u(x)\,dx = Lu^*$ and $0 \le \chi_0 \le \overline{X}$ satisfies

$$\|u\|_{C^{2+\theta}([0,L])},\ \|v\|_{C^{2+\theta}([0,L])} \le C. \tag{4.13}$$

Proof. (1) The strict inequalities involving the minimum and maximum follow from continuity, nonconstancy, and the prescribed average. The strict L^2 inequality follows from the Cauchy–Schwarz inequality, since equality would require u to be constant.

(2) Integration of the first stationary equation gives

$$u_x = \chi_0 \frac{u^m}{(1+v)^\beta} v_x. \tag{4.14}$$

The proof then follows the same arguments as in Lemma 4.1(2), with $a = b = \alpha = 0$, which show that u, u_x, v, v_x, and v_{xx} are uniformly bounded. Then by a priori estimates for elliptic equations, there are $\theta \in (0,1)$ and $C = C(\overline{X}, u^*, \mu, \nu, \beta, \gamma, L, m) > 0$ such that (4.13) holds. □

4.1 Proof of Theorem 1.3(1)

In this subsection, we give the outline of the proof of Theorem 1.3(1).

Outline of the proof of Theorem 1.3(1).

Let (u^*, v^*) be a positive constant stationary solution of (1.3). Let X and $\mathcal{A} = \mathcal{A}(u^*)$ be as in (1.18) and (1.19), respectively. Let

$$Y := \begin{cases} L^2((0,L)) & \text{if} \quad a, b > 0 \\ \\ \{u \in L^2((0,L)) \,|\, \int_0^L u(x)dx = 0\} & \text{if} \quad a = b = 0 \end{cases} \tag{4.15}$$

with norm $\|u\| = \|u\|_{L^2((0,L))}$. Let

$$\mathcal{B} = \mathcal{B}(u^*) := \{u \in X \,|\, u + u^* \in \mathcal{A}\}. \tag{4.16}$$

For $w \in \mathcal{B}$, let $v(w)$ solve

$$\begin{cases} -v_{xx} + \mu v = \nu(w+u^*)^\gamma, & x \in (0,L) \\ v_x(0) = v_x(L) = 0, \end{cases} \tag{4.17}$$

and set

$$F(\chi_0, w) = w_{xx} - \chi_0 \Big(\frac{(w+u^*)^m (v(w))_x}{(1+v(w))^\beta} \Big)_x + a(w+u^*) - b(w+u^*)^{1+\alpha}.$$

Then F is a mapping from $(0,\infty) \times \mathcal{B} \to Y$. If $(\chi_0, w) \in (0,\infty) \times \mathcal{B}$ satisfies $F(\chi_0, w) = 0$, then $(u, v) = (w + u^*, v(w))$ is a positive stationary solution of (1.3), where $v(w)$ solves (4.17).

We prove Theorem 1.3(1) by applying the global bifurcation theorem of Shi and Wang [26, Theorem 4.4] to the mapping $F : (0,\infty) \times \mathcal{B} \subset (0,\infty) \times X \to Y$.

The set $\mathcal{B}$ is open and convex in X. The operator F is smooth, and its linearizations and their convex interpolations are compact perturbations of the Neumann Laplacian, hence Fredholm of index zero. At $p^* = (\chi^*, 0)$, the Fourier representation gives

$$\ker D_w F(\chi^*, 0) = \operatorname{span}\{\phi_{n_0}\}, \qquad D_{\chi_0 w} F(\chi^*, 0)\phi_{n_0} = \alpha_{n_0}\phi_{n_0} \notin \operatorname{range}\big(D_w F(\chi^*, 0)\big),$$

since $\alpha_{n_0} > 0$ and the range is orthogonal to ϕ_{n_0}. The domain spaces have Hilbert norms, so the remaining smoothness hypotheses of [26, Theorem 4.4] also hold.

Hence, [26, Theorem 4.4] yields that the connected sets $\mathcal{C}^{\pm}_{n_0}$ stated in Theorem 1.3(1) satisfy alternatives (i)–(iii). Moreover, it is clear that

$$\mathcal{S} = \{(\chi_0, u^* + w) \,|\, \chi_0 \in (0,\infty),\ w \in \mathcal{B} \setminus \{0\},\ F(\chi_0, w) = 0\}.$$

Since $\mathcal{C}$ is a connected component of $\overline{\mathcal{S}} \subset (0,\infty)\times\mathcal{A}$ (see (1.20)) (the closure is taken in $\mathbb{R}\times\mathcal{A}$), it is relatively closed in $(0,\infty)\times\mathcal{A}$. By local uniqueness of the bifurcating curve, after decreasing ε if necessary, $\mathcal{G}^{\mp}_{n_0}$ is relatively open in $\mathcal{C}$. It follows that $\mathcal{C}\setminus\mathcal{G}^{\mp}_{n_0}$ is relatively closed in $(0,\infty)\times\mathcal{A}$. Therefore, its connected component $\mathcal{C}^{\pm}_{n_0}$ containing $\mathcal{G}^{\pm}_{n_0}$ is relatively closed in $(0,\infty)\times\mathcal{A}$. Moreover, $\mathcal{G}^{\pm}_{n_0}\cup\{(\chi^*,u^*)\}$ is a connected subset of $\mathcal{C}\setminus\mathcal{G}^{\mp}_{n_0}$. Hence, $(\chi^*,u^*)\in\mathcal{C}^{\pm}_{n_0}$. □

4.2 Proof of Theorem 1.3(2)

Proof of Theorem 1.3(2). Suppose $\mathcal{C}^{+}_{n_0}$ is not compact but its χ_0-coordinate is bounded. Choose a sequence of nonconstant stationary solutions $(\chi_j,u_j)=(\chi_j,w_j+u^*)\in\mathcal{C}^{+}_{n_0}$ with no convergent subsequence in $\mathcal{C}^{+}_{n_0}$. Without loss of generality, we may assume that

$$0<\chi_j\le\overline{X},\qquad \chi_j\to\bar\chi\ge 0$$

for some $\overline{X}>0$ and $\bar\chi\ge 0$. By Lemma 4.1(2) and regularity for elliptic equations, we have

$$\sup_j\left(\|u_j\|_{C^{2+\theta}}+\|v_j\|_{C^{2+\theta}}\right)<\infty$$

for some $\theta>0$. Hence, after extraction and relabeling, we may assume that

$$u_j\to\bar u,\qquad v_j\to\bar v\quad\text{in } C^2([0,L]).$$

The limit is a nonnegative stationary solution at $\chi_0=\bar\chi$.

By the relative closedness of $\mathcal{C}^{+}_{n_0}$ in $(0,\infty)\times\mathcal{A}$, $(\bar\chi,\bar u)$ lies on the boundary of $(0,\infty)\times\mathcal{A}$. By Lemma 4.1(2), $(u_j(x),v_j(x))$ converges to $(\bar u(x),\bar v(x))$ in $C^2([0,L])$ and $(\bar u(x),\bar v(x))$ is a nonnegative stationary solution of (1.3) with $\chi_0=\bar\chi$. By Lemma 4.1(1), $\sup_{x\in[0,L]}\bar u(x)\ge u^*$. Then strong comparison principle for elliptic equations implies that $\bar u(x)>0$ for all $x\in[0,L]$. Thus $(\bar\chi,\bar u)$ belongs to the closure of $\mathcal{S}$ in $\mathbb{R}\times\mathcal{A}$. By (1.20), $\bar\chi>0$. Hence, $(\bar\chi,\bar u)\in(0,\infty)\times\mathcal{A}$, contradicting the boundary conclusion above. Therefore, if $\mathcal{C}^{+}_{n_0}$ is not compact, then its χ_0-coordinate is unbounded. The same argument applies to $\mathcal{C}^{-}_{n_0}$. □

4.3 Proof of Theorem 1.3(3)

Proof of Theorem 1.3(3). First, we prove $u^+(\chi_0)$ is monotone for χ_0 near χ^*. From (2.46), set $s=\sqrt{\frac{\alpha_1}{\beta_1}\tilde\chi}+\mathcal{O}(|\tilde\chi|)$, and write the positive local half-branch as

$$u^+=u^*+s\cos\left(\tfrac{\pi x}{L}\right)+r_s,\qquad s>0,\qquad \|r_s\|_{C^2}=o(s).$$

Write $u_s=u^+$, and denote its sensitivity and signal by χ_s and v_s. The signal equation gives $v_s=v^*+s\eta\phi_1+o(s)$ in C^2, where $\eta=\nu\gamma(u^*)^{\gamma-1}/(\mu+\lambda_1)$. Since $\chi_s\to\chi^*$ and $\chi^*(u^*)^m\eta/(1+v^*)^\beta=1$, differentiating (4.14) gives $u_s''=s\phi_1''+o(s)$ uniformly. Together with $H^2\hookrightarrow C^1$, this yields $\|r_s\|_{C^2}=o(s)$. The Neumann conditions imply $r_s'(0)=r_s'(L)=0$. Integrating r_s'' from either endpoint yields

$$|r_s'(x)|\le\|r_s''\|_\infty\min\{x,L-x\}.$$

Since $\sin\left(\frac{\pi x}{L}\right)\ge\frac{2}{L}\min\{x,L-x\}$, we obtain

$$(u^+)'(x)\le\left(-\frac{2\pi s}{L^2}+\|r_s''\|_\infty\right)\min\{x,L-x\}<0\qquad(0<x<L)$$

for sufficiently small $s > 0$. Although $(u^+)'(0) = (u^+)'(L) = 0$, for every $0 \le x_1 < x_2 \le L$,

$$u^+(x_2) - u^+(x_1) = \int_{x_1}^{x_2} (u^+)'(x)\,dx < 0.$$

Thus u^+ is strictly decreasing on $[0, L]$. Similarly, u^- is strictly increasing on $[0, L]$.

Next, we prove all the solutions in $\mathcal{C}_1^{\pm}$ are monotone. By the first equation in (1.3) with $a = b = 0$ and $u_t = 0$, we have

$$u_x = \frac{\chi_0 u^m}{(1+v)^\beta} y, \qquad y_{xx} + \Big(\frac{\chi_0 \nu\gamma u^{m+\gamma-1}}{(1+v)^\beta} - \mu\Big) y = 0, \qquad y = v_x, \quad y(0) = y(L) = 0. \tag{4.18}$$

For a nonconstant solution, uniqueness implies

$$y(x_0) = y_x(x_0) = 0 \quad \Longrightarrow \quad y \equiv 0 \quad \Longrightarrow \quad u \equiv u^*,$$

a contradiction. Thus, every zero of $v_x(x)$ is simple, which implies that every zero of $u_x(x)$ is simple. Therefore, strict monotonicity persists under sufficiently small perturbations among stationary solutions.

Suppose that a sequence of decreasing nonconstant solutions satisfies $(\chi_j, u_j) \to (\widehat{\chi}, u^*)$ in $(0, \infty) \times \mathcal{A}$. If $D_w F(\widehat{\chi}, 0)$ were invertible, the implicit function theorem would exclude such a sequence. Hence, $\widehat{\chi} = \chi_n(u^*)$ for some $n \ge 1$, where (1.9) with $a = 0$ gives strictly increasing critical values. At each such value the kernel is $\operatorname{span}\{\phi_n\}$ and the parameter crossing is transversal, as follows from (1.6). The local simple-eigenvalue bifurcation theorem [26, Theorem 4.3] applied to the stationary mapping F therefore gives

$$u_j = u^* + s_j \phi_n + r_j, \qquad s_j \neq 0, \quad \|r_j\|_{H^2} = o(|s_j|).$$

If $n \ge 2$, the derivative of ϕ_n takes both signs in $(0, L)$. Since $H^2(0, L) \hookrightarrow C^1([0, L])$, the derivative of u_j would also take both signs for large j, a contradiction. Thus, $n = 1$ and $\widehat{\chi} = \chi^*$.

Consequently,

$$\{(\chi^*, u^*)\} \cup \big\{(\chi_0, u) \in \mathcal{C}_1^+ : u_x < 0 \text{ on } (0, L)\big\}$$

is relatively open and closed in $\mathcal{C}_1^+$.

Near a positive stationary limit, H^2 convergence bootstraps to C^2 convergence. A nonconstant limit of decreasing solutions remains strictly decreasing by uniqueness in (4.18). The interior sign and simple endpoint zeros persist under small stationary perturbations. Constant limits occur only at (χ^*, u^*); there, local uniqueness and removal of the opposite half-arc retain the decreasing orientation.

Connectedness, and the analogous argument for $\mathcal{C}_1^-$, yield

$$u_x < 0 \text{ on } \mathcal{C}_1^+ \setminus \{(\chi^*, u^*)\}, \qquad u_x > 0 \text{ on } \mathcal{C}_1^- \setminus \{(\chi^*, u^*)\},$$

with the inequalities understood on $(0, L)$. Thus, alternative (ii) is excluded. Moreover,

$$\int_0^L (u - u^*)\phi_1\,dx = -\frac{L}{\pi} \int_0^L u_x \sin\big(\frac{\pi x}{L}\big)\,dx \neq 0,$$

excluding alternative (iii). Hence, both connected sets are noncompact.

Suppose that sensitivity is bounded on $\mathcal{C}_1^+$. Since this component is noncompact, choose a sequence of nonconstant solutions $(\chi_j, u_j) \in \mathcal{C}_1^+$ with no convergent subsequence in $\mathcal{C}_1^+$. Lemma 4.2(2) gives, after extraction,

$$\chi_j \to \bar{\chi} \ge 0, \qquad (u_j, v_j) \to (\bar{u}, \bar{v}) \quad \text{in } C^2([0, L]).$$

By the relative closedness of $\mathcal{C}_1^+$ in $(0,\infty)\times\mathcal{A}$, $(\bar\chi,\bar u)$ lies on the boundary of $(0,\infty)\times\mathcal{A}$. By Lemma 4.2(2), $(u_j(x),v_j(x))$ converges to $(\bar u(x),\bar v(x))$ in $C^2([0,L])$ and $(\bar u(x),\bar v(x))$ is a nonnegative stationary solution of (1.3) with $\chi_0=\bar\chi$. Note that $\int_0^L \bar u(x)\,dx = Lu^*$. Hence, $\bar u\not\equiv 0$ and strong comparison principle for elliptic equations implies that $\bar u(x)>0$ for all $x\in[0,L]$. Thus, $(\bar\chi,\bar u)$ belongs to the closure of $\mathcal{S}$ in $\mathbb{R}\times\mathcal{A}$. By (1.20), $\bar\chi>0$. Hence, $(\bar\chi,\bar u)\in(0,\infty)\times\mathcal{A}$, contradicting the boundary conclusion above. Hence (1.21) holds. The same argument applies to $\mathcal{C}_1^-$.

Finally, adapting the arguments of [30, Theorem 3.1] and [21, Theorem 3.1], we now prove that the mass of u concentrates at $x=0$ along $\mathcal{C}_1^+$ and at $x=L$ along $\mathcal{C}_1^-$.

Let $(\chi_j,u_j)\in\mathcal{C}_1^+$ with $\chi_j\to\infty$, and let v_j be the corresponding signal. Then u_j is strictly decreasing and $\int_0^L u_j\,dx = Lu^*$. The v-estimate in the proof of Lemma 4.2 yields

$$\|v_j\|_\infty \le V := \frac{\nu}{\mu}(u^*)^\gamma + 2\nu L^2(u^*)^\gamma,$$

independently of j and χ_j. By (4.18), the function $w_j=-(v_j)_x$ is positive on $(0,L)$ and satisfies

$$-w_j'' = \Big(\frac{\chi_j\nu\gamma u_j^{m+\gamma-1}}{(1+v_j)^\beta}-\mu\Big)w_j.$$

We claim that $u_j(\delta)\to 0$ for every $\delta\in(0,L)$. Otherwise, after passing to a subsequence, $u_j(\delta)\ge\varepsilon$ for some $\varepsilon>0$. Monotonicity then implies that, on $I=[\delta/2,\delta]$,

$$\frac{\chi_j\nu\gamma u_j^{m+\gamma-1}}{(1+v_j)^\beta}-\mu \ge \frac{\chi_j\nu\gamma \varepsilon^{m+\gamma-1}}{(1+V)^\beta}-\mu \longrightarrow\infty \qquad \text{uniformly.}$$

Set $\psi(x)=\sin\left(\frac{2\pi(x-\delta/2)}{\delta}\right)$, and $\lambda_I=\frac{4\pi^2}{\delta^2}$. Integration by parts gives

$$\begin{aligned}\int_{\delta/2}^{\delta}\Big(\frac{\chi_j\nu\gamma u_j^{m+\gamma-1}}{(1+v_j)^\beta}-\mu\Big)w_j\psi\,dx &= \lambda_I\int_{\delta/2}^{\delta} w_j\psi\,dx + w_j(\delta)\psi'(\delta)-w_j(\delta/2)\psi'(\delta/2)\\ &< \lambda_I\int_{\delta/2}^{\delta} w_j\psi\,dx,\end{aligned}$$

which is impossible, since the coefficient of w_j exceeds λ_I throughout I for large j. Thus, $u_j(\delta)\to 0$, and

$$\int_\delta^L u_j(x)\,dx \le (L-\delta)u_j(\delta)\longrightarrow 0.$$

Together with the fixed mass, this implies

$$\int_0^L f(x)u_j(x)\,dx\longrightarrow Lu^*f(0) \qquad \text{for every } f\in C([0,L]).$$

Hence, $u_j(x)\,dx$ converges weakly-* to $Lu^*\delta_0$ as finite measures on $[0,L]$. By reflection, the corresponding limit along $\mathcal{C}_1^-$ is $Lu^*\delta_L$. In particular, $\|u_j\|_\infty\to\infty$ along either branch. □

5 Spike-like stationary solutions and proof of Theorem 1.4

In this section, we prove Theorem 1.4.

5.1 Constant limits at large sensitivity

We use a common form of the convergence argument in [19, Theorem 4.3, arXiv version 1].

Lemma 5.1. *Fix all model parameters except χ_0. Let (u_n, v_n) be positive stationary solutions of (1.3) with $\chi_0 = \chi_n \to \infty$. Assume either $a, b > 0$, with $u^* = (a/b)^{1/\alpha}$, or $a = b = 0$ and $\int_0^L u_n\,dx = Lu^* > 0$. Suppose $\sup_n \|u_n\|_{L^r(0,L)} < \infty$ for some finite $r > \max\{1, \gamma\}$, and put $q = r/\gamma > 1$. There exist a subsequence $\{n_k\}$ and a constant $c \geq 0$ such that*

$$u_{n_k}^\gamma \rightharpoonup c \quad \text{in} \quad L^q, \quad v_{n_k} \to \frac{\nu}{\mu} c \quad \text{in} \quad C^1, \quad v_{n_k} \rightharpoonup \frac{\nu}{\mu} c \quad \text{in} \quad W^{2,q}. \tag{5.1}$$

Proof. First, by the boundedness assumption of $\{u_n\}$ in L^r, $\{v_n\}$ is bounded in $W^{2,q}$. By weak compactness and the compact embedding $W^{2,q}(0,L) \hookrightarrow C^1([0,L])$, without loss of generality, we may assume that there are nonnegative functions W, V such that

$$u_n^\gamma \rightharpoonup W \text{ in } L^q, \qquad v_n \to V \text{ in } C^1, \qquad v_n \rightharpoonup V \text{ in } W^{2,q}.$$

Note that

$$-V_{xx} + \mu V = \nu W, \qquad V_x(0) = V_x(L) = 0. \tag{5.2}$$

Also, $(1 + v_n(x))^{-\beta} \geq \kappa > 0$ for some constant $\kappa > 0$ and all $x \in [0, L]$. In both models, Lemma 4.1(1) or fixed mass gives $\int_0^L u_n\,dx \leq M := Lu^*$. For every $m > 0$, direct integration of the stationary u–equation from 0 to x, using the Neumann conditions at 0, gives

$$(u_n)_x = \chi_n \frac{u_n^m}{(1+v_n)^\beta}(v_n)_x - F_n, \tag{5.3}$$

where

$$F_n(x) = \int_0^x \left(a u_n(y) - b u_n(y)^{1+\alpha}\right)\,dy, \qquad \|F_n\|_\infty \leq 2aM.$$

Note that $F_n = 0$ in the minimal case.

Next, we show that $V(\cdot)$ is a constant function. Assume that $V(\cdot)$ is not a constant function. Choose a non-empty closed interval $J \subset \{V_x \neq 0\} \cap (0, L)$ and $0 \leq \varphi \in C_c^1(0, L)$ equal to one on J, supported where V_x has a fixed nonzero sign. Then for all large n, $(v_n)_x$ has that sign and $|(v_n)_x| \geq \sigma > 0$ for some $\sigma > 0$ and x on the support. By (5.3), we have

$$\chi_n \kappa \sigma \int_J u_n^m\,dx \leq \left| -\int_0^L u_n \varphi_x\,dx + \int_0^L F_n \varphi\,dx \right| \leq M\|\varphi_x\|_\infty + 2aM\|\varphi\|_1.$$

Thus, $\int_J u_n^m\,dx \to 0$. With $H = \sup_n \|u_n\|_r$, Hölder's and Chebyshev's inequalities yield, for every $\varepsilon > 0$,

$$\int_J u_n^\gamma\,dx \leq \varepsilon^\gamma |J| + H^\gamma \left(\varepsilon^{-m} \int_J u_n^m\,dx\right)^{1-\gamma/r}.$$

Letting $n \to \infty$ and then $\varepsilon \to 0$ shows that $W = 0$ almost everywhere on J. Exhausting the components of $\{V_x \neq 0\}$ gives $WV_x = 0$ almost everywhere. Multiplication of (5.2) by V_x therefore gives

$$\partial_x(V_x^2 - \mu V^2) = 0 \quad \text{almost everywhere.}$$

Note that $V_x^2 - \mu V^2$ is absolutely continuous and hence constant. At the extrema of V, including the Neumann endpoints, $V_x = 0$. Thus, $(\min V)^2 = (\max V)^2$, and nonnegativity makes V constant, which contradicts the assumption that V is not a constant function. Equation (5.2) then gives $W = c$ and $V = \nu c/\mu$. □

5.2 Proof of Theorem 1.4

Proof. We first prove (1.22) for both models at once.

To this end, choose $x_n \in [0, L]$ such that $u_n(x_n) = \max_{[0,L]} u_n$. Extract a further subsequence with $x_n \to x^* \in [0, L]$. Assume that (1.22) does not hold with this x^*. Then, after passing to a further subsequence if necessary, we may assume that there is $\delta > 0$ such that

$$|(u_n)_x| \le K \quad \text{on } I \quad \text{for} \quad n \gg 1, \tag{5.4}$$

where $I = [x^* - \delta, x^* + \delta] \cap [0, L]$. Note that, for all large n, $x_n \in I$, and for any $x \in I$,

$$\max_{[0,L]} u_n = u_n(x_n) = u_n(x) + \int_x^{x_n} (u_n)_x(y)dy.$$

It then follows that

$$\max_{[0,L]} u_n \le \frac{1}{|I|}\int_I u_n\, dx + K|I| \le \frac{Lu^*}{|I|} + K|I| \quad \forall\, n \gg 1. \tag{5.5}$$

Consequently, for every $p > 1$,

$$\limsup_{n\to\infty} \int_0^L u_n^p(x)dx < \infty. \tag{5.6}$$

By the assumption $\liminf_{n\to\infty} \max_{[0,L]} u_n > u^*$ in the theorem, we may assume that

$$h_n := u_n(x_n) = \max_{[0,L]} u_n \longrightarrow \ell > u^*. \tag{5.7}$$

Fix $r > \max\{1, \gamma\}$ and set $q = r/\gamma > 1$. The uniform L^∞ bound implies boundedness in $L^r(0, L)$. By Lemma 5.1, there exist a subsequence $\{n_k\}$ and a constant $c \ge 0$ such that

$$u_{n_k}^\gamma \rightharpoonup c \ \text{ in } \ L^q, \quad v_{n_k} \to \frac{\nu}{\mu}c \ \text{ in } \ C^1, \quad v_{n_k} \rightharpoonup \frac{\nu}{\mu}c \ \text{ in } \ W^{2,q}. \tag{5.8}$$

By (5.4) and (5.5), we may assume that $u_{n_k}(x) \to U(x)$ uniformly on I as $k \to \infty$ for some $U \in C(I)$. Since $u_n^\gamma \rightharpoonup c$, we have $U^\gamma = c$ on I. Moreover, $x_n \to x^*$ and $u_n(x_n) \to \ell$ imply $U(x^*) = \ell$, so $c = \ell^\gamma$. It then follows that

$$0 \le \int_0^L (h_{n_k}^\gamma - u_{n_k}^\gamma)\, dx \longrightarrow L\ell^\gamma - Lc = 0.$$

Thus, $u_{n_k}^\gamma \to \ell^\gamma$ in L^1. This implies that $L\ell = \lim_{n_k} \int_0^L u_{n_k}\, dx \le Lu^*$, which contradicts (5.7) in either model. Therefore, along the subsequence with $x_n \to x^*$, (1.22) holds with this x^*.

Next, we prove (1.23). By Lemma 4.1,

$$\limsup_{n\to\infty} \int_0^L u_n^{1+\alpha}(x)dx < \infty.$$

Then by the assumption $\gamma < 1 + \alpha$ and Lemma 5.1, (5.8) holds with $q = \frac{1+\alpha}{\gamma}$. This together with (4.1) and Hölder's inequality implies that

$$cL = \lim_{n_k\to\infty} \int_0^L u_{n_k}^\gamma(x)dx \le L(u^*)^\gamma.$$

(1.23) then follows. The further extraction preserves (1.22), since the local gradient norms tend to infinity; relabeling gives the same subsequence for both conclusions. □

6 Numerical analysis of the bifurcation and spike-like solutions

In this section, we numerically investigate bifurcation and the formation of spike-like profiles in the minimal and non-minimal models. We first describe the spatial discretization and prove that the discrete instability threshold converges to its continuous counterpart as the mesh is refined. We then present numerical experiments for parameters in the super-critical and sub-critical regimes, comparing the finite-time evolution near the instability threshold and examining the profiles at larger values of χ_0.

6.1 Numerical scheme and discrete threshold on a mesh and proof of Proposition 1.2

In this subsection, we describe the finite-difference schemes for the two models. For the non-minimal model, we discretize the space variable in (1.3) by a finite-difference method on N uniform subintervals, with $N+1$ grid points, and obtain the following discretized system of ODEs:

$$\begin{cases} u_i' = (\Delta_h u)_i + \beta\chi_0 \dfrac{u_i^m}{(1+v_i)^{\beta+1}}(D_h v_i)^2 - m\chi_0 \dfrac{u_i^{m-1}}{(1+v_i)^{\beta}}(D_h u_i)(D_h v_i) \\ \qquad\quad - \chi_0 \dfrac{u_i^m}{(1+v_i)^{\beta}}\left(\mu v_i - \nu u_i^{\gamma}\right) + a u_i - b u_i^{1+\alpha}, \quad i = 0,1,\dots,N, \\ 0 = (\Delta_h v)_i - \mu v_i + \nu u_i^{\gamma}, \quad i = 0,1,\dots,N, \\ u_{-1} = u_1,\ u_{N+1} = u_{N-1},\ v_{-1} = v_1,\ v_{N+1} = v_{N-1}. \end{cases} \tag{6.1}$$

Here $x_i = ih$, $h = L/N$, $u_i(t) \approx u(t,x_i)$, and $v_i(t) \approx v(t,x_i)$. For $1 \le i \le N-1$, set

$$D_h w_i := \frac{w_{i+1} - w_{i-1}}{2h}, \qquad (\Delta_h w)_i := \frac{w_{i+1} - 2w_i + w_{i-1}}{h^2},$$

while the ghost-point convention gives

$$D_h w_0 = D_h w_N = 0, \qquad (\Delta_h w)_0 = \frac{2(w_1 - w_0)}{h^2}, \qquad (\Delta_h w)_N = \frac{2(w_{N-1} - w_N)}{h^2}.$$

The vector $(u_i^*, v_i^*) = (u^*, v^*)$, $0 \le i \le N$, is a constant solution of (6.1).

For the minimal model, we use the same grid and elliptic solver, but its constant equilibria are parametrized by the conserved mean. We therefore compare stability within one fixed-mass class. Changing the prescribed mean changes the reference constant equilibrium; the neutral constant mode is excluded from the instability calculation. The continuous u–equation preserves mass because it is in divergence form and the Neumann conditions eliminate the boundary flux. The expanded pointwise scheme (6.1) has no exact discrete product rule and need not preserve that identity. Therefore, for the minimal experiments we instead discretize the complete diffusion–chemotaxis flux. Set

$$q_i := \frac{u_i^m}{(1+v_i)^{\beta}}, \qquad \mathcal{F}_{i+\frac{1}{2}} := \frac{u_{i+1} - u_i}{h} - \chi_0 \frac{q_i + q_{i+1}}{2}\frac{v_{i+1} - v_i}{h}.$$

We use

$$u_i' = \begin{cases} 2\mathcal{F}_{\frac{1}{2}}/h, & i = 0, \\ \left(\mathcal{F}_{i+\frac{1}{2}} - \mathcal{F}_{i-\frac{1}{2}}\right)/h, & 1 \le i \le N-1, \\ -2\mathcal{F}_{N-\frac{1}{2}}/h, & i = N. \end{cases} \tag{6.2}$$

The endpoint nodes in (6.2) represent half control volumes, so the trapezoidal discrete mass

$$M_h(u) \coloneqq h\left(\frac{u_0}{2} + \sum_{i=1}^{N-1} u_i + \frac{u_N}{2}\right)$$

satisfies $dM_h(u)/dt = 0$ by telescoping. We work at fixed mass and use mean-zero perturbations

$$u_0(x) = u^* \pm \varepsilon\, \phi_{n_0}(x),$$

so that $\frac{1}{L}\int_0^L u_0(x)\,dx = u^*$.

As mentioned in the introduction, $\chi_N^{*,\mathrm{disc}}(u^*,\beta)$ given in (1.26) is the instability threshold of the constant solution (u^*, v^*) of (6.1). For the minimal model, the same threshold applies to the conservative scheme (6.2) on the fixed-mass subspace: at a constant state, its linearization is $\Delta_h\delta u - \chi_0 q^* \Delta_h \delta v$, with $q^* = (u^*)^m/(1+v^*)^\beta$, which agrees with the expanded scheme when $a = b = 0$. In the following, we prove Proposition 1.2, that is, $\lim_{N\to\infty} \chi_N^{*,\mathrm{disc}}(u^*,\beta) = \chi^*_{a,b,\beta}(u^*)$.

Proof of Proposition 1.2. Recall that discrete Neumann eigenvalues $\lambda_{n,N}^{\mathrm{disc}}$ are given by (1.24). For every fixed $n \geq 1$, with $N \geq n$,

$$\lambda_{n,N}^{\mathrm{disc}} \longrightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2 \quad \text{as} \quad N \to \infty.$$

Moreover, since $\sin x \geq 2x/\pi$ for $0 \leq x \leq \pi/2$, $\lambda_{n,N}^{\mathrm{disc}} \geq \frac{4n^2}{L^2}$.

Let

$$F(\lambda) \coloneqq \frac{(1+v^*)^\beta}{\nu\gamma(u^*)^{m+\gamma-1}} \frac{(\lambda + a\alpha)(\mu+\lambda)}{\lambda}. \tag{6.3}$$

and $C \coloneqq \frac{(1+v^*)^\beta}{\nu\gamma(u^*)^{m+\gamma-1}}$. Then $F(\lambda) \geq C\lambda$. On the other hand, $\sup_{N\geq 1} F(\lambda_{1,N}^{\mathrm{disc}}) < \infty$. Consequently, there exists $K \geq 1$, independent of N, such that no mode $n > K$ can minimize either the discrete or the continuum thresholds. Thus, for all sufficiently large N,

$$\chi_N^{*,\mathrm{disc}}(u^*,\beta) = \min_{1\leq n\leq K} F\left(\lambda_{n,N}^{\mathrm{disc}}\right), \qquad \chi^*_{a,b,\beta}(u^*) = \min_{1\leq n\leq K} F(\lambda_n).$$

Since convergence on this finite set of modes is uniform, (1.27) then follows. □

6.2 Numerical experiments and observations

In this subsection, we use these schemes to illustrate the super-critical and sub-critical cases in both models. We compare the near-threshold computations with the local theory and examine the profiles at the reported final times for larger sensitivities.

In Figures 6.1–6.4, the value denoted by χ^* in each plot title is the discrete threshold $\chi_N^{*,\mathrm{disc}}$ for the mesh used in that experiment. The first rows of Figures 6.1 and 6.2 show near-threshold experiments for the non-minimal model with $\beta = 2$ and $\beta = 3$, respectively. These choices correspond to the first block in the left panel (non-minimal model) of Table 3.1 and lie on opposite sides of the transition $\beta_+ \approx 2.1586$. Thus, Figure 6.1 uses parameters in the super-critical regime, whereas Figure 6.2 uses parameters in the sub-critical regime.

The first rows of Figures 6.3 and 6.4 show near-threshold experiments for the minimal model with $\beta = 4$ and $\beta = 5$, respectively. These choices correspond to the first block in the right

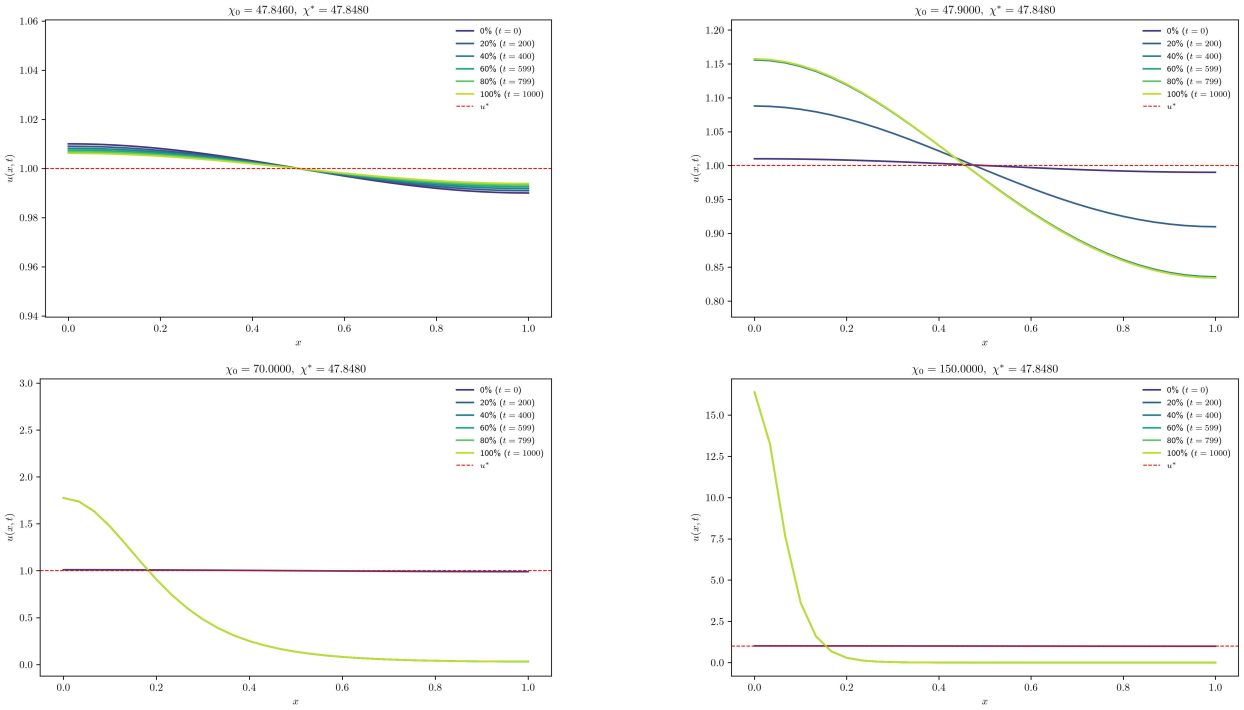


Figure 6.1: Numerical experiments for the parameters $a = b = \alpha = m = \mu = \nu = \gamma = L = 1$, $\beta = 2$, $\varepsilon = 0.01$, mesh size $N = 30$, and final time $t = 1000$, $u_0 = 1 + 0.01\cos(\pi x)$, $\chi_0 \in \{47.846,\ 47.9,\ 70,\ 150\}$, and $\chi_{30}^{*,\mathrm{disc}} = 47.8480$.

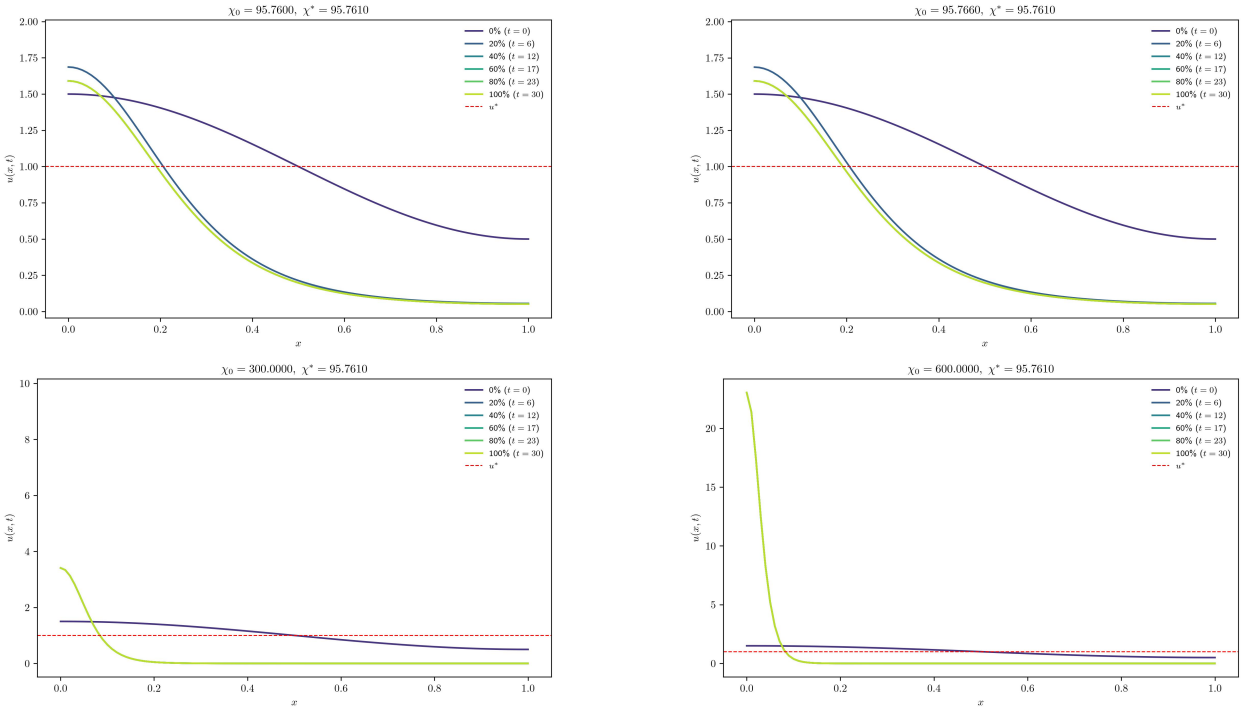


Figure 6.2: Numerical experiments for the parameters $a = b = \alpha = m = \mu = \nu = \gamma = L = 1$, $\beta = 3$, $\varepsilon = 0.5$, mesh size $N = 100$, and final time $t = 30$, $u_0 = 1 + 0.5\cos(\pi x)$, and $\chi_0 \in \{95.76,\ 95.766,\ 300,\ 600\}$, and $\chi_{100}^{*,\mathrm{disc}} = 95.7610$.

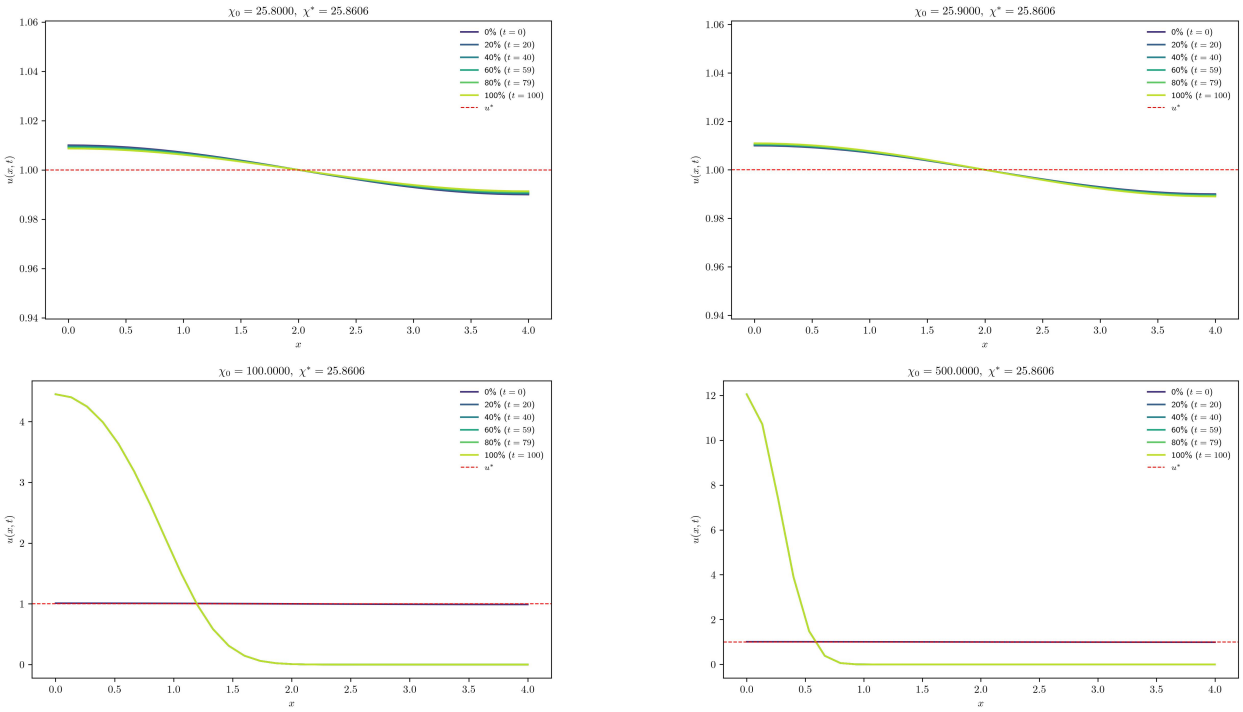


Figure 6.3: Numerical experiments for the parameters $a = b = 0$, $u^* = 1$, $\alpha = m = \mu = \nu = \gamma = 1$, $L = 4$, $\beta = 4$, $\varepsilon = 0.01$, mesh size $N = 30$, and final time $t = 100$, $u_0 = 1 + 0.01\cos(\frac{\pi x}{4})$, and $\chi_0 \in \{25.8,\ 25.9,\ 100,\ 500\}$, and $\chi_{30}^{*,\text{disc}} = 25.8606$.

panel (minimal model) of Table 3.1 and lie on opposite sides of the transition $\beta_+ \approx 4.3692$. Thus, Figure 6.3 uses parameters in the super-critical regime, whereas Figure 6.4 uses parameters in the sub-critical regime.

From these numerical experiments, we observe the following:

- Nonconstant profiles persist as χ_0 increases, which suggests that the local bifurcation branches for χ_0 near the threshold χ^* extend to $\chi_0 \to \infty$, consistent with the unbounded alternative in Theorem 1.3(1).
- The maxima of the nonconstant profiles stay above u^* in the tested regimes, where $u^* = (a/b)^{1/\alpha}$ for the non-minimal model and is the prescribed average for the minimal model, which suggests that there are non-constant stationary solutions satisfying the conditions in Theorem 1.4, and then spike-like solutions exist as $\chi_0 \to \infty$.

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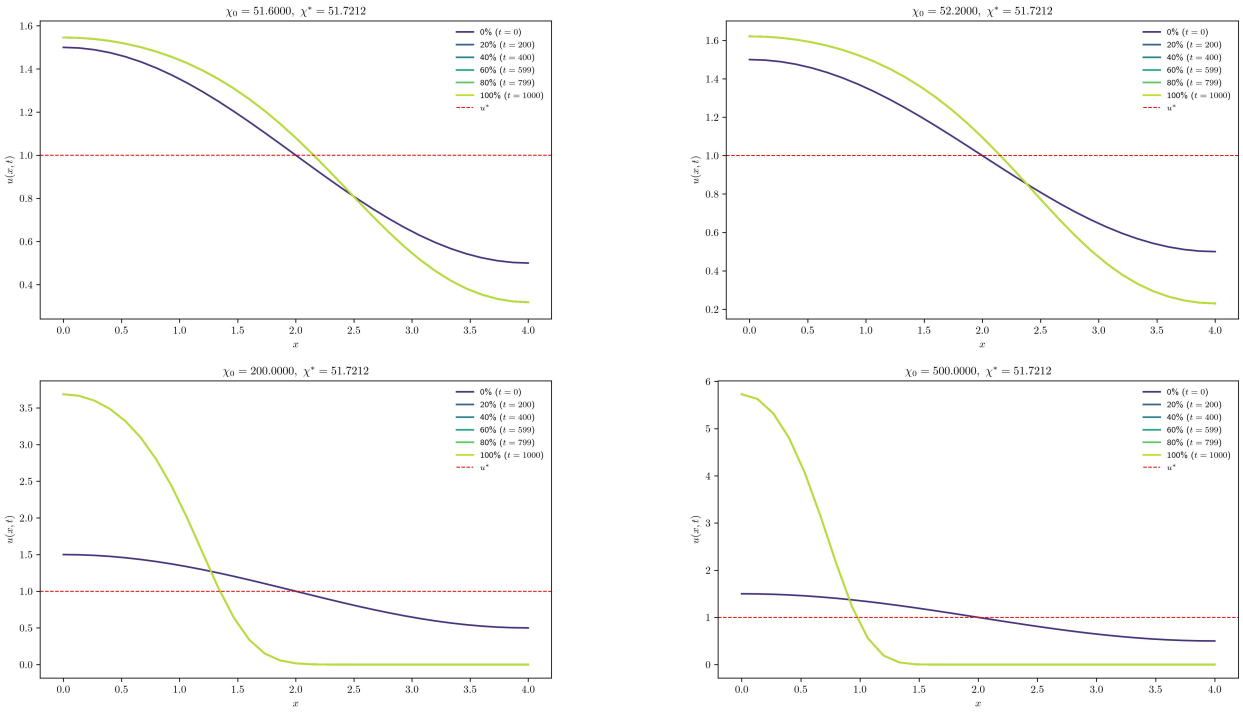


Figure 6.4: Numerical experiments for the parameters $a = b = 0$, $u^* = 1$, $\alpha = m = \mu = \nu = \gamma = 1$, $L = 4$, $\beta = 5$, $\varepsilon = 0.5$, mesh size $N = 30$, and final time $t = 1000$, $u_0 = 1 + 0.5\cos(\frac{\pi x}{4})$, and $\chi_0 \in \{51.6,\ 52.2,\ 200,\ 500\}$, and $\chi_{30}^{*,\text{disc}} = 51.7212$.

Data and code availability

The code and figures for the numerical experiments are available at `https://github.com/ianruau/Chemotaxis_simulation`. All experiments were performed using `simulation.py`. The script `paper_3_numerical_experiments.sh` reproduces the numerical experiments and generates the figures stored in `paper_iii_numerical_experiments` that were used in this paper.

AI declaration

The authors manually derived all the expressions involved in the proofs of Theorems 1.1 and 1.2 in the paper. After the authors' manual computation of the expressions in (1.16), (2.44), (2.45), and (3.3), OpenAI's GPT models were used to assess that our hand-derived expressions matched the models' expressions.

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